

NEURAL BELLMAN OPERATORS

QIAN QI

Peking University

We develop Neural Bellman Operators with policy-specific payoff acceptance and separate unrestricted optimal comparison. In the original stopped consumption–portfolio economy, newly trained networks generate stochastic consumption and a nonzero, exactly self-financing portfolio. A continuum-mixture theorem and a verified Jensen-gap bound establish actual policy improvement at every initial state of a rectangle, rather than only at four expert states. Five confirmatory seeds attain final regret below .002909 there, and initial-to-final true regret falls by more than 85.8 percent. Fixed-policy price transport extends a .003327 bound over the rectangle and an adjustment-price interval. An additional .002 of initial wealth is sufficient compensation for each learned stochastic policy. A direct same-family comparator is more accurate at every vertex and materially cheaper to generate. The network observes time only; restricted wealth feedback arises through its analytic decoder. Global feedback accuracy, localized optimality, policy improvement, and high-dimensional warm-start effects are kept distinct. The full-domain .01 objective and all prior substantive results are retained.

KEYWORDS: Stochastic control, Neural Bellman operators, Preference adjustment, Validated numerics, Policy verification, Computational economics.

1. INTRODUCTION

A useful numerical solution of a dynamic economic model must improve decisions, not merely the function used to evaluate them. This distinction is particularly important when a neural policy and a value witness are fitted jointly. A smaller upper bound on policy loss need not mean that the policy has improved: the same policy may receive a better witness. We therefore separate the generation of a policy, the verification of its own payoff, and comparison with the unrestricted optimum.

This paper develops Neural Bellman Operators with that separation. The original consumption–portfolio economy permits costly adjustment of preferences, correlated preference and financial shocks, and stopping at the boundary of the economic state space. Its primitives, admissible controls, stopping rule, and accuracy objective are unchanged. The new solver generates genuinely stochastic consumption and a nonzero portfolio. Neural coefficients determine an adapted consumption rule; an explicit contingent-claim price finances the rule and determines its portfolio. Independently enclosed stopped payoffs, rather than a changing critic, decide whether a proposed policy is accepted.

There are three substantive additions. First, we give a continuous-time self-financing and admissibility theorem for the executed neural policies. The decoder is not a discretized wealth recursion: its conditional-price identity holds between decision dates and for every realization of the traded factor. Second, we prove a continuum transfer theorem for four expert policies. Concavity of running utility alone is not enough to order two mixtures, so we also bound the old mixture’s Jensen gain. This supplies an independently verified improvement in actual policy value at every initial state in a rectangle. Third, we combine policy-specific lower bounds with the previously verified unrestricted dual upper bound. This proves useful neural accuracy without requiring the inaccurate historical feedback critic to serve as the evaluator.

Five confirmatory seeds, four experts per seed, and four retained work levels were fixed before the R20 execution. All sixty noninitial expert proposals pass a policy-specific strict

acceptance test before further training. The present revision supplies their missing manuscript, the continuum arguments, a price-sensitivity theorem, and independent re-execution of the frozen-policy checks. It does not relabel those R20 optimization runs as new random R21 experiments. At time zero and adjustment price $k = 2$, every final neural mixture has regret below .002909 uniformly on

$$K = [1.98, 2.02] \times [1.24, 1.26].$$

The final policies improve actual payoff over their corresponding initial mixtures by at least 0.01762319 throughout K . Consequently, true regret is reduced by more than 85.8 percent at every point of this rectangle. A further, explicitly post-training calculation keeps the policies fixed and proves a regret bound of 0.00332694 throughout $K \times [1.5, 2.5]$.

The comparison is not made easy for the neural method. A direct, non-neural stochastic-control transcription is evaluated with the same four-state checker, unrestricted optimal upper bound, and continuum theorem. It reaches the same .005 tolerance and is faster in the recorded generation-plus-verification ledger. The five neural policies nevertheless have payoff lower bounds strictly above the policy-specific upper bounds of each of three named deterministic time-control transcriptions, uniformly on K . This identifies an economically consequential stochastic policy improvement without asserting superiority over all conventional solvers. A budget-financed wealth increment of .002 suffices for each learned mixture to attain at least the original optimum's upper bound, at the uncompensated initial wealth, throughout K .

The domain of each assertion matters. A theorem uniform on K at time zero is not a theorem for all initial states of D and all starting times. The outstanding all-domain feedback target remains .01. Neither the localized dual comparison nor the new pricing decoder is presented as a globally sharp continuation supersolution. The earlier continuation-trace results, complete-domain failures, signed residual decompositions, actor-critic interventions, and nonlinear high-dimensional experiments remain in the paper and supplement. They are indispensable controls on the interpretation of the new results, rather than observations to remove after a successful localized experiment.

2. THE STOPPED PREFERENCE-ADJUSTMENT ECONOMY

Time is continuous, with horizon $T = 1$. Preferences u and wealth x evolve as

$$du_s = \theta_s ds + .05 dZ_s^u, \quad (1)$$

$$dx_s = \{(.02 + .06p_s)x_s - c_s\} ds + .2p_s x_s dZ_s^x, \quad (2)$$

where $d\langle Z^u, Z^x \rangle_s = -.25 ds$. The state domain is

$$D = (1.2, 2.8) \times (.5, 2), \quad A = [.05, .8] \times [-.2, .2] \times [-.5, .8].$$

An admissible control $a = (c, \theta, p)$ is progressively measurable, takes values in A , and is used up to the first exit from D or time one, whichever occurs first. The controls in the original optimization are not restricted to deterministic time functions or to neural networks.

For adjustment price $k \in [.5, 8]$, the running payoff and settlement are

$$\ell_k(u, c, \theta) = \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2, \quad (3)$$

$$G(u, x) = -.02(u-2)^2 + .1 \log x, \quad g(s, u, x) = G(u, x) - 8(1-s). \quad (4)$$

Let $\tau = 1 \wedge \inf\{s \geq t : (u_s, x_s) \notin D\}$ and $\rho = .04$. The criterion is

$$J_k(t, z; a) = \mathbb{E}_{t,z} \left[\int_t^\tau e^{-\rho(s-t)} \ell_k(u_s, c_s, \theta_s) ds + e^{-\rho(\tau-t)} g(\tau, u_\tau, x_\tau) \right]. \quad (5)$$

Here $V_k = \sup_{a \in \mathcal{A}} J_k(\cdot; a)$. The liquidation fee makes a stopping-trace error economically meaningful even when the training distribution assigns little mass near an exit. It is therefore included explicitly in verification rather than approximated by an average boundary loss.

For a smooth test function v , define the discounted residual

$$\begin{aligned} \mathcal{R}_a v = & v_t + \theta v_u + \{(.02 + .06p)x - c\}v_x + .00125v_{uu} + .02p^2x^2v_{xx} - .0025pxv_{ux} \\ & + \frac{c^{1-u}}{1-u} - \frac{k}{2}\theta^2 - .04v. \end{aligned} \quad (6)$$

All constants in the certificate denote the displayed exact decimals. Training uses floating-point approximations of them, but the certification program encloses the exact constants and treats stored binary64 network weights as exact real numbers.

The central initial condition is $(t, u, x) = (0, 2, 1.25)$. We also study the nondegenerate rectangle

$$K = [1.98, 2.02] \times [1.24, 1.26]. \quad (7)$$

The claim on K is uniform over its interior, not a statement about 72 isolated initial conditions. Its proof is given in Section 15, and its tolerance differs from the sharper central-state tolerance.

3. CONSTRUCTIVE NEURAL VERIFICATION

We first state the verification result for a class of stopped controlled diffusions. Let Z take values in a bounded domain, with drift b_a , covariance matrix Σ_a , running reward f_a , constant discount $\rho \geq 0$, and settlement g on the parabolic boundary. Write

$$\mathcal{R}_a v = v_t + b_a \cdot Dv + \frac{1}{2} \Sigma_a : D^2 v + f_a - \rho v.$$

The assumptions below concern the stochastic model and the computed witness. They do not assume regularity of the unknown optimal value.

ASSUMPTION 1: *For every admissible control, the stopped diffusion and its payoff are well defined. The controlled coefficients and payoffs are bounded on the stopped domain. The candidate feedback π is admissible. The witness v is continuously differentiable in time and twice continuously differentiable in state on a neighborhood of the closed domain, and its derivatives are bounded there. Discounted Itô's formula and optional stopping hold for these bounded stopped quantities.*

The compact economy in Section 2 with the stored smooth, bounded neural feedback satisfies these requirements. Boundedness also permits localization of the stochastic integral before taking expectations.

THEOREM 1—A complete residual and improvement certificate: *Under Assumption 1, suppose verified nonnegative functions $e(s), q(s)$ and a number $b \geq 0$ satisfy, at every state and*

time,

$$|\mathcal{R}_\pi v| \leq e(s), \quad 0 \leq \sup_{a \in A} \mathcal{R}_a v - \mathcal{R}_\pi v \leq q(s), \quad |v - g| \leq b \text{ on the stopping boundary.} \quad (8)$$

Then, for every initial state z and time t ,

$$0 \leq V(t, z) - J(t, z; \pi) \leq 2b + \int_t^T e^{-\rho(s-t)} \{2e(s) + q(s)\} ds. \quad (9)$$

The same conclusion applies to any admissible implementation whose action lies in a verified output-error neighborhood of the stored feedback, provided that the residual and improvement inequalities have been enclosed over that neighborhood.

The theorem is deliberately an a posteriori statement. It is useful because every item in (8) has a finite implementation, not because its stochastic comparison identity is new. A training history can propose the witness and policy, but cannot replace any of the inequalities. The proof in Appendix B separates the policy-evaluation and upper-comparison errors; each stopping-trace discrepancy is paid once.

3.1. A finite implementation

Fix a rational rectangular cover of the entire state–time domain, including every stopping face. On each cell, interval forward differentiation gives the value, first derivatives, and relevant second derivatives of the stored network. An action oracle bounds the maximum over the continuous action set. Interval evaluation of the deployed policy gives its residual and the difference from this global maximum. Taking the largest bounds in each time slab produces e and q . Facewise interval evaluation produces b . Finally, integration of these piecewise-constant bounds yields (9).

A cell is accepted only if all operations have finite, valid enclosures and every potential action branch is covered. No failed or inconclusive cell can be removed from the domain. Refinement replaces a cell by a complete cover of that cell, while keeping previous results in the execution record. The method either returns the certified inequality for the specified stored object or records that a requested accuracy has not been established at the available resolution. This is a stopping criterion for an accuracy claim, not a criterion based on validation loss.

In the economy, the action maximization separates into three scalar problems conditional on the interval-enclosed derivatives. Consumption is strictly concave, and its maximizer is

$$c^* = \text{proj}_{[.05, .8]}(v_x^{-1/u}) \quad \text{when } v_x > 0, \quad c^* = .8 \quad \text{when } v_x \leq 0.$$

The preference action is $\theta^* = \text{proj}_{[-.2, .2]}(v_u/k)$. The portfolio objective is

$$h_1 p + \frac{1}{2} h_2 p^2, \quad h_1 = .06 x v_x - .0025 x v_{ux}, \quad h_2 = .04 x^2 v_{xx}.$$

A negative curvature gives a clipped vertex; nonnegative curvature gives an endpoint maximum. When an interval does not determine the sign, the calculation encloses the whole feasible portfolio interval. Thus uncertainty enlarges the bound instead of excluding an action. These formulas preserve the original continuous control opportunities.

3.2. Transport across model coefficients

The affine dependence on k used later is not a general argument for perturbing dynamics. For the latter we use residual transport. Consider another model with the same state domain, action set, and settlement, but possibly different coefficients and discount. Let its discount be at least $\underline{\rho} \geq 0$, and let $\tilde{\mathcal{R}}_a$ denote its residual.

PROPOSITION 1—Coefficient transport: *Suppose the original checks (8) hold, and interval coefficient bounds verify*

$$\sup_{a \in A} |\tilde{\mathcal{R}}_a v - \mathcal{R}_a v| \leq d(s) \quad (10)$$

throughout the common domain. If Assumption 1 also holds for the new model, its policy regret is at most

$$2b + \int_t^T e^{-\underline{\rho}(s-t)} \{2e(s) + q(s) + 2d(s)\} ds. \quad (11)$$

If the settlement changes by at most b_{par} , replace b by $b + b_{\text{par}}$.

For example, the modulus in (10) can be evaluated directly from

$$|(\tilde{b}_a - b_a) \cdot Dv| + \frac{1}{2} |(\tilde{\Sigma}_a - \Sigma_a) : D^2 v| + |\tilde{f}_a - f_a| + |\tilde{\rho} - \rho| |v|.$$

It includes the change in the covariance matrix rather than treating volatility perturbations as drift changes. The proposition does not couple paths or equate stopping times across economies. Each model is verified against its own stopped dynamics on the common domain. The executed coefficient box is reported in Section 22.

4. STOCHASTIC NEURAL POLICIES WITH SEPARATE PAYOFF ACCEPTANCE

The economy and notation are as specified above. In this section $r = .02$, the risky asset has excess drift .06 and volatility .2, and $Y_0 = 1$ with

$$dY_t = .02Y_t dt - .3Y_t dW_t^x.$$

The equivalent martingale measure satisfies $dW_t^{x,Q} = dW_t^x + .3dt$. Thus $\log Y_t$ has mean $-.025t$ under the physical measure and $.065t$ under the martingale measure, in both cases with variance $.09t$.

4.1. Neural generation and exact deployment semantics

A 1–16–16–3 network with two tanh hidden layers takes only the normalized time-slab coordinate and produces coefficients for sixteen equal time slabs. Current preference and the traded factor are not network inputs; factor dependence enters the fixed decoder. The budget offset is differentiated inside the proposal objective. For slab j , the deployed policy is

$$c_j(y) = .5 + .3 \sigma(b_j - s_j \log y), \quad .1 \leq s_j \leq 6.5, \quad 0 \leq \theta_j \leq .2, \quad (12)$$

where $\sigma(a) = (1 + e^{-a})^{-1}$. The stored dyadic coefficients, not an unspecified real-number neural forward pass, define the delivered policy. The network is compiled once; the contingent-claim decoder below then acts on the observable factor. Each expert has 355 neural parameters. Four

experts have 1,420 parameters, apart from the decoder. The network is trained from random weights without inherited policy or value labels. Its accuracy is established by checking its delivered policy, not by assuming convergence of Adam.

For a given initial state (u_i, x_i) , let

$$R_i = e^r \left(x_i - \int_0^1 e^{-rt} \mathbb{E}^Q[c_i(t, Y_t)] dt \right). \quad (13)$$

The directed price check verifies $.5 < R_i < .5001$. In particular, R_i is an exact price-defined reserve enclosed by an interval; it is not an approximately balanced budget. Define

$$F_i(t, y) = R_i e^{-r(1-t)} + \int_t^1 e^{-r(s-t)} \mathbb{E}^Q[c_i(s, Y_s) | Y_t = y] ds, \quad (14)$$

$$X_t^i = F_i(t, Y_t), \quad p_i(t, y) = -1.5 y \partial_y F_i(t, y) / F_i(t, y). \quad (15)$$

Consumption and preference drift may jump at the deterministic slab dates; wealth is continuous. The price is smooth inside each slab and has matching values at its endpoints.

THEOREM 2—Self-financing stochastic decoder: *Every policy satisfying (12)–(13) is a progressively measurable admissible policy of the original economy. Its wealth satisfies*

$$dX_t = ((.02 + .06p_t)X_t - c_t) dt + .2p_t X_t dW_t^x.$$

For all traded-factor realizations, before preference stopping,

$$.5 < X_t < 1.282253 < 2, \quad 0 \leq p_t < .706888 < .8.$$

There is no wealth-boundary exit. The preference process and its stopping rule remain those of the original model.

The proof uses the conditional-price equation, not a simulated or time-discretized budget. It also checks the portfolio bound globally in $y > 0$. Thus the nonzero portfolio is an actual admissible control, not a diagnostic evaluated only on sampled paths.

4.2. Strict acceptance of policy values

Let $[L_i^n, U_i^n]$ enclose the stopped expected payoff of expert i at retained work level n . A proposed expert is accepted only if

$$L_i^{n+1} > U_i^n. \quad (16)$$

The checker completes before further proposals. Rejection restores both the network and Adam state to the incumbent. The original experiment retains every proposal, certificate, decision, network, compiled coefficients, and cost. It contains no learned critic in the acceptance test. Therefore (16) proves a strict increase in that expert's actual payoff at its specified initial state. It does not assert global optimization of the neural parameters.

The optimal comparator is separate. Let $\bar{V}_k(z)$ be the inherited unrestricted dual upper bound on the original stopped economy. A policy can be accurate because $\bar{V}_k(z) - L_k^\pi(z)$ is small, but merely reducing $\bar{V}_k - L_k^\pi$ is never substituted for (16). The source identities and costs of the dual library are retained separately from neural generation.

4.3. A policy on a continuum of initial states

Order the four vertices $z_i = (u_i, x_i)$ of K lexicographically. For $z \in K$, let $\lambda_i(z)$ be its bilinear interpolation weights. On the same Brownian paths, run the four subaccounts and set

$$\begin{aligned} u_t &= \sum_i \lambda_i u_t^i, & X_t &= \sum_i \lambda_i X_t^i, & c_t &= \sum_i \lambda_i c_t^i, \\ \theta_t &= \sum_i \lambda_i \theta_t^i, & p_t &= \frac{\sum_i \lambda_i p_t^i X_t^i}{X_t}. \end{aligned} \quad (17)$$

Initial wealth is exactly $x = \sum_i \lambda_i x_i$. The wealth-weighted portfolio remains in the original admissible interval. This is an adapted policy with stored initial-state context and observable subaccounts, not a claim that a single map of current (t, u, x) alone represents every expert history.

THEOREM 3—Uniform optimality from verified expert payoffs: *Let $p_E = 2 \exp\{-.58^2/(2 \cdot .0025)\}$. For any fixed $k \in [.5, 8]$, the mixture (17) satisfies*

$$J_k^{\text{mix}}(z) \geq \sum_i \lambda_i(z) L_{i,k} - 16p_E, \quad z \in K. \quad (18)$$

If $\bar{V}_k$ is convex in z on K , then

$$0 \leq V_k^*(z) - J_k^{\text{mix}}(z) \leq \max_i \{\bar{V}_k(z_i) - L_{i,k}\} + 16p_E. \quad (19)$$

The explicit dual upper used here is convex: its state dependence is affine, the support function of an interval slope, and a nonnegative exponential localization term. Convexity of the unknown value function is not assumed.

The theorem uses joint concavity of $f(u, c) = c^{1-u}/(1-u)$ on $[1.2, 2.8] \times [.5, .8]$, concavity of settlement, and the common Brownian coupling. Stopping is dealt with by the explicit event correction, rather than by an invalid assertion that arbitrary stopped payoffs are globally concave.

4.4. Why expert improvement does not suffice for mixture improvement

Even if all four experts improve, the old mixture might have a larger Jensen benefit than the new mixture. Ordering the four corners alone therefore does not prove a policy improvement at every interior state. We address this additional identification problem directly.

THEOREM 4—Policy-sensitive continuum improvement: *Suppose Γ_0 verifies*

$$J_2^{\text{mix},0}(z) \leq \sum_i \lambda_i(z) U_i^0 + \Gamma_0, \quad z \in K.$$

Then the final mixture has uniform payoff improvement at least

$$d = \min_i (L_i^f - U_i^0) - \Gamma_0 - 16p_E. \quad (20)$$

When $d > 0$ and (19) gives final regret at most B , true regret contracts pointwise by

$$\frac{V_2^*(z) - J_2^{\text{mix},f}(z)}{V_2^*(z) - J_2^{\text{mix},0}(z)} \leq \frac{B}{B+d}, \quad z \in K. \quad (21)$$

For each executed initial four-expert system, the common slopes and preference drifts allow the directed Hessian construction in the appendix to compute Γ_0 . This construction yields $d \geq 0.01762319$ and $B/(B+d) \leq 0.14110003$ for all five seeds.

We distinguish the sixty strict expert transitions, the four work-level accuracy bounds on K , and the proved initial-to-final mixture improvement. We do not assert that every consecutive mixed policy improves throughout K : the small later expert gains do not by themselves establish that stronger statement.

4.5. A fixed-policy price-sensitivity theorem

For a fixed policy, neither the state law nor the stopping time changes with k . Write

$$B_i^{\text{full}} = \frac{1}{2} \int_0^1 e^{-.04t} (\theta_t^i)^2 dt.$$

Then $B_i^{\text{stop}} \in [B_i^{\text{full}} - .02C_{\rho E}, B_i^{\text{full}}]$ and

$$J_{i,k} = J_{i,2} - (k-2)B_i^{\text{stop}}. \quad (22)$$

Use this interval identity with the fixed final neural coefficients; do not retrain or interpolate neural solutions in k . The unrestricted dual upper is bounded between its verified price nodes by convex chords. On each price segment, an affine payoff lower bound subtracted from the affine dual chord is affine at each vertex. Its maximum is therefore bounded by its endpoints. Applying Theorem 3 at those endpoints proves, without a sampled price grid,

$$\sup_{z \in K, k \in [1.5, 2.5]} \{V_k^*(z) - J_k^{\text{mix},f}(z)\} \leq 0.00332694 < .005 \quad (23)$$

for every seed. This is a post-training sensitivity theorem about the fixed policies, not an additional confirmatory training experiment.

5. EXECUTED NEURAL ACCURACY AND MATCHED LOCAL COMPARISONS

The confirmatory design fixes seeds 20100–20104, the four vertices of K , Adam step size .005, and retained budgets 0, 100, 400, 1000. The exploratory seed 20000 is excluded. The independent checker encloses physical-measure utility, preference costs, settlement, martingale-measure consumption prices, Gaussian and stopping tails, and the global control bounds. Eighty neural expert checkpoints were originally checked; all twenty final experts were also checked by a separate MPFR arithmetic implementation. R21 replays the frozen-policy checks and uses the hull of the available enclosures for the reported final bounds. The shared mathematics is not thereby an independent formal proof.

The initial policies miss .01. Every seed reaches both .01 and .005 at the first retained noninitial budget of 100 updates. All five final bounds are below .003. None reaches the additional .002 reporting threshold. The last column is an actual-payoff improvement bound, obtained from Theorem 4; it is not the change in an optimality upper bound. The final arithmetic hull gives bounds between approximately 0.00289021 and 0.00290848. Table entries for error bounds are rounded upward, and gain lower bounds downward.

TABLE I

FOUR-EXPERT NEURAL POLICIES: BOUNDS UNIFORM ON K , AT $t = 0, k = 2$. FINAL BOUNDS USE THE HULL OF BOTH ARITHMETIC IMPLEMENTATIONS.

Seed	0 updates	100	400	1,000	Gain over initial
20100	0.03819183	0.00363672	0.00299180	0.00290848	0.01840213
20101	0.03760120	0.00417052	0.00301777	0.00290261	0.01785789
20102	0.03732944	0.00384199	0.00301067	0.00289514	0.01762319
20103	0.03755641	0.00385873	0.00300813	0.00289022	0.01783956
20104	0.03778334	0.00364572	0.00299527	0.00289318	0.01803023

5.1. The neural and classical cost frontier

The conventional stochastic comparator directly optimizes sixteen triples of coefficients in (12), with no hidden neural layers. It uses L-BFGS-B, the same conditional-price decoder, the same four initial states, and precisely the same payoff and dual checks. The retained budgets are 0, 10, 50, 200, with convergence recorded when earlier. This is a matched *local* continuous-time accuracy comparison, not a surrogate comparison on a different discrete economy.

The direct stochastic method reaches .005 at ten retained iterations for each expert and has final K -uniform bound 0.00283871. At the first .005 hit, neural seed 20100 uses 1.107 seconds for proposal generation and 15.036 seconds for the accumulated binary interval checks, or 16.143 seconds in total after outward rounding. The direct method uses .153 and 15.360 seconds, respectively, or 15.512 seconds. All five seed-specific ledgers, target hits and misses, and final wall times appear in the supplement. Different optimizer iterations are not equated: the comparison is made at the common checked tolerance.

These are recorded costs conditional on access to the common optimal upper. A cold-start ledger additionally charges the full 415.789 seconds of the inherited twenty-node dual-library construction, or explicitly declares its reuse. The final twenty-expert MPFR audit and R21 replays are additional publication-validation costs, not silently free checks. Recorded neural peak process memory is retained. The R20 direct-stochastic run did not record peak RSS; its 192 scalar coefficients must not be substituted for process memory. Consequently, the data support a local accuracy frontier and an explicit cost ledger, but neither a complete global MC/SL frontier nor a neural memory or speed advantage.

5.2. A policy-sensitive stochastic improvement

Three separately executed deterministic SLSQP transcriptions, with 16, 32, and 64 time slabs, receive their own supporting affine payoff upper bounds on K . These are upper bounds for the named policies, not for the optimal value. The neural mixtures' lower bounds exceed each such affine upper by at least 0.00455889 throughout K . This comparison orders actual stopped payoffs without changing the evaluator to favor the neural policy. It establishes an improvement over these three deterministic policies; the sharper and faster direct stochastic comparator remains visible and precludes a blanket neural-superiority claim.

5.3. Budget-feasible welfare compensation for the learned policies

For every final neural expert, retain the learned slopes and preference drifts and raise only its consumption intercept by a common budget offset. Finance the resulting consumption stream from initial wealth $x_i + .002$, and verify the exact residual reserve again using (13). The twenty compensated-policy lower bounds exceed the original, uncompensated optimal-value upper

bounds by at least 0.00004475 after continuum transfer. Thus each learned mixture, provided with an additional .002 of initial wealth, achieves at least $V_2^*(u, x)$ for every $(u, x) \in K$. The sufficient compensation is .16 percent at reference wealth 1.25, and less than .162 percent throughout K . It is neither the minimum compensating variation nor an external unfunded transfer. Unlike the historical .0078 result for a deterministic library, this statement concerns the new learned stochastic policies.

5.4. Relation to the historical whole-domain experiments

The R18 complete-domain feedback results remain 7.278–7.344 against the unchanged .01 target. Their final policy-side residual component is zero for all ten accessibility-aware seeds; the remaining bound is the positive comparison component. The all-seed actor–critic factorial, signed decompositions and trace-floor trajectories are retained below and in the supplement. A smaller historical bound cannot identify actor progress.

The new evaluator does not apply another interval wrapper to those failed actors or critics. It evaluates newly trained stochastic consumption policies and uses a separate localized unrestricted dual upper. Its success on K is therefore compatible with the historical fixed-critic floor above 6.27. It does not prove that the historical critic has learned the correct continuation trace, or supply a sharp supersolution on the whole cylinder. The retained one-sided continuation targets provide constructive information for that separate global problem.

The nonlinear high-dimensional control material is likewise preserved with its actual attribution. The initialization-independent strongly-convex theorem is not evidence that training is necessary. The subsequent learned-gradient theorem and 32-query matched-tolerance experiment do identify a limited reduction in online gradient work, including 31 of 32 paired queries at dimension 128 and tolerance .01. They do not establish a same-hardware amortized speedup, performance for all cube states from those 32 queries, or a broad stochastic high-dimensional advantage.

6. STATE-FEEDBACK REALIZATION AND THE REPRESENTATION ABLATION

The same-family comparison is a test of the neural parameterization itself, not merely another external baseline. To interpret that test correctly, we specify both the information observed by the network and the state feedback implemented by its decoder.

The network input is only the normalized time-slab coordinate. Neither the current preference nor the traded factor is a network input. The factor enters the prescribed logistic consumption rule; its self-financing portfolio is supplied by conditional pricing. Within a seed, the random-number generator is reset before each of the four experts, and their initial parameter dictionaries are exactly identical. Different seeds give five different initial dictionaries. The four objectives and budget offsets differ. Thus there are five independent seed replicates, not twenty independently initialized experts. The exact equality of the initial slopes and effort vectors used in the Jensen proof is consistent with this design.

PROPOSITION 2—Restricted Markov realization: *Fix an expert or fixed initial-state mixture, and denote its consumption, reserve, and conditional price by c_λ , R_λ , and F_λ . Put $A_r(t) = (1 - e^{-r(1-t)})/r$. For every $t < 1$, the map $y \mapsto F_\lambda(t, y)$ is continuously differentiable and strictly decreasing from*

$$H_\lambda(t) = R_\lambda e^{-r(1-t)} + .8A_r(t) \quad \text{to} \quad L_\lambda(t) = R_\lambda e^{-r(1-t)} + .5A_r(t).$$

It has an inverse $y_\lambda(t, x)$ on this open wealth interval. Before stopping, the deployed policy consequently has the equivalent feedback representation

$$\hat{c}_\lambda(t, x) = c_\lambda(t, y_\lambda(t, x)), \quad \hat{p}_\lambda(t, x) = -1.5 y_\lambda(t, x) \frac{\partial_y F_\lambda(t, y_\lambda(t, x))}{x}, \quad \hat{\theta}_\lambda(t) = \theta_\lambda(t).$$

This is a restricted wealth-feedback policy indexed by its fixed initial-state context. It is not a single trained network on all current states (t, u, x) .

PROOF: Every slab slope is strictly positive. For any positive future interval, logistic consumption is strictly decreasing in its factor argument and has limits .8 and .5. Conditional factors multiply the present factor by a strictly positive lognormal random variable. Differentiation under the bounded integrand gives $\partial_y F_i < 0$ for $t < 1$; the derivative is continuous at every positive y . Dominated convergence gives the two displayed limits. Finite convex combinations preserve strict monotonicity and these endpoint formulas, since the mixture reserve is $R_\lambda = \sum_i \lambda_i R_i$. The inverse-function theorem applies. Substitution of $X_t = F_\lambda(t, Y_t)$ proves equality of the feedback and subaccount implementations along every pre-stopping path. At $t = 1$ the interval collapses to the reserve and no inverse is required. The portfolio formula agrees with the wealth-weighted mixture in (17). Q.E.D.

The proposition identifies real feedback in the delivered controls without attributing it to an unobserved neural input. It does not execute a full-state neural Bellman solve, remove the restricted policy class, or add reactions to current preference shocks. The unrestricted value in the optimality bounds is nevertheless the original economy’s value, not the optimum of that restricted class.

TABLE II

REPRESENTATION ABLATION AT THE FINAL RETAINED WORK LEVEL. BOTH METHODS USE THE SAME STOCHASTIC LOGISTIC FAMILY AND UNRESTRICTED UPPER. THE FINAL COLUMN IS A CERTIFIED PAYOFF ORDERING, NOT MERELY A COMPARISON OF REGRET BOUNDS.

u	x	Best neural bound	Direct bound	Direct payoff gain $\geq$
1.98	1.24	0.00261425	0.00258714	0.00002670
1.98	1.26	0.00289022	0.00283871	0.00005111
2.02	1.24	0.00268964	0.00266465	0.00002459
2.02	1.26	0.00260024	0.00255442	0.00004542

The direct parameterization has a smaller final certified regret at every vertex than every neural seed. A further interval subtraction, using the hulls of the independently recomputed payoff intervals, establishes a strict *actual payoff* advantage at those vertices; the last column reports its lower bound. This is adverse evidence for the hidden-layer parameterization in the tested regime. The vertex ordering is not asserted for every interior mixture: different mixtures may have different Jensen gains. The direct mixture does have the sharper proved uniform regret bound shown earlier.

Pooling the common verifier cost would obscure the generation result. The direct final four-expert solve takes approximately 1.14 seconds of generation, against 10.63–10.79 seconds for a neural seed. Both are charged for all retained checkpoints. Neural training is the reported generation work, not an omitted extra offline stage. The shared dual construction, final MPFR pass, and R21 frozen-policy replay are separately charged as explained above. Missing direct process peak memory remains unreported, rather than being imputed from parameter storage.

TABLE III

UNPOOLED FOUR-EXPERT COST AT THE FINAL RETAINED WORK LEVEL (SECONDS). THE FINAL-CHECK COLUMN IS A SUBSET OF ALL CHECKPOINT CHECKING, NOT AN ADDITIONAL CHARGE. THE SHARED DUAL AND MPFR COSTS REMAIN SEPARATE.

Method	Generation	All checking	Final checking
Seed 20100	10.631	30.002	7.509
Seed 20101	10.730	30.197	7.610
Seed 20102	10.788	30.245	7.597
Seed 20103	10.627	30.174	7.577
Seed 20104	10.638	30.242	7.560
Direct	1.143	30.597	7.598

The full-state target is unchanged. The best inherited complete-domain feedback certificate remains 7.278319, and the earlier R19 bound over K is .011816625 despite its sub-.01 reference-state result. The new sub-.003 uniform result concerns the explicitly defined stochastic mixture and the new proof, not a revision of either historical observation. The current experiment therefore establishes useful neural-generated policy accuracy and policy improvement on K , but does not establish a neural representation advantage or global full-state NBO accuracy.

7. RETAINED SOLVER THEORY AND EXPERIMENTAL DEVELOPMENT

The following sections preserve the substantive R14–R19 mathematical and numerical development. Their experiment-specific bounds concern the objects named in those sections; they are not relabeled as R21 training. In particular, deterministic price libraries, historical full-state feedback networks, and the new stochastic neural mixtures remain separate policies. The new uniform results above do not revise the old observations by deletion.

8. POLICY-VALUE-SEPARATED NEURAL UPDATES

For a policy π and initial state z , write $L_\pi(z) \leq J_\pi(z) \leq U_\pi(z)$ for independently established payoff enclosures. Write $\bar{V}(z) \geq V(z)$ for an optimal upper comparison. The two upper bounds have different meanings: U_π evaluates one policy, whereas $\bar{V}$ ranges over every admissible policy. A better $\bar{V}$ can sharpen a certificate without improving π .

PROPOSITION 3—Separated acceptance and observable progress: *Let the preceding bounds hold on a set K . A proposal π' satisfying*

$$\inf_{z \in K} \{L_{\pi'}(z) - U_\pi(z)\} > 0 \quad (24)$$

strictly improves the true payoff at every $z \in K$. More strongly, if, for some $0 < \kappa < 1$,

$$L_{\pi'} - U_\pi \geq \kappa(\bar{V} - L_\pi) \quad \text{on } K, \quad (25)$$

then $V - J_{\pi'} \leq (1 - \kappa)(V - J_\pi)$ on K . If instead a verified gain $J_{\pi'} - J_\pi \geq d > 0$ and final regret bound $V - J_{\pi'} \leq B$ hold uniformly, then

$$\frac{V - J_{\pi'}}{V - J_\pi} \leq \frac{B}{B + d}. \quad (26)$$

All inequalities compare economic payoffs. An increase in L_π alone does not imply improvement: both the policy and the width of its evaluation interval can change. Equation (25) is a checkable sufficient progress condition, not an assumption that Adam always produces such a proposal. Its numerical use below is an a posteriori certificate for retained transitions, not an a priori optimization rate.

The evaluator need not be neural. For example, a smooth fixed-policy evaluation function q obeys the stopped identity

$$J_\pi(z) = q(z) + \mathbb{E}_z^\pi \int_0^\tau e^{-\rho s} \mathcal{R}_\pi q(s, Z_s) ds + \mathbb{E}_z^\pi [e^{-\rho \tau} (g - q)(\tau, Z_\tau)].$$

Separate upper and lower enclosures of these two terms yield L_π, U_π . A distinct supersolution, relaxation, or monotone comparison can produce $\bar{V}$. These roles cannot be interchanged merely because both calculations involve a value function.

8.1. The executed neural policy block

The new actor has two hidden tanh layers of width 16 and 338 trainable parameters. It maps the midpoints of 16 time slabs to two outputs N_c, N_θ . Let

$$w_j = \int_{j/16}^{(j+1)/16} e^{-\rho s} ds, \quad Q_r = \sum_j w_j, \quad b = 1.25 - .5e^{-\rho \cdot 0.2} - 10^{-7}.$$

At the reference wealth its mathematical controls are

$$c_j = \frac{b}{Q_r} + .006 \left\{ \tanh N_c(t_j) - \frac{\sum_i w_i \tanh N_c(t_i)}{Q_r} \right\}, \quad (27)$$

$$\theta_j = .1 + .1 \tanh N_\theta(t_j), \quad p_j = 0.$$

The exact budget equation implies terminal wealth $.5 + 10^{-7}e^{-\rho \cdot 0.2}$. For an initial wealth $x \in [1.24, 1.26]$, add $(x - 1.25)/Q_r$ to every consumption slab. Terminal wealth is then unchanged. The implementation serializes the compiled binary64 controls, checks their actual budget and action bounds by directed arithmetic, and treats these exact dyadic controls as the delivered policy. It does not assume that floating-point network compilation is exact.

The initial-state map is held fixed after initialization. Thus this experiment is a neural-generated time-control subclass of the original adapted policy set, not the state-feedback architecture of the complete-cover study. The upper bound still concerns the unrestricted original problem. Keeping these two facts together permits a genuine optimality certificate for the delivered neural policy without asserting a whole-domain feedback result.

Seeds 19100–19104, budgets 0, 50, 200, 800, learning rate .003, and the payoff gate were declared before execution. Full-batch proposal quadrature uses 16 time nodes and 40 Gaussian nodes per slab. It evaluates the original CRRRA expectation, adjustment cost, and terminal reward; it does not use an inherited policy or an optimal-value label. This ordinary floating-point objective generates proposals only. At each retained checkpoint, the independent rational-Gauss evaluator, with Cauchy remainders, normal-tail bounds, and an original-stopping correction, establishes the policy-value interval. The final reproduction executes this value gate before the next Adam update; rejected proposals would restore the incumbent network and optimizer state. Every prescribed proposal is retained. All fifteen noninitial proposals pass the strict gate at the reference state.

8.2. Transferring policy improvement to initial states

Separate Lipschitz bounds for two values can overwhelm a small but economically meaningful improvement. We instead control their difference under common shocks. Let a, b denote two feasible slab policies with the same wealth-transfer rule. On a common refinement of their time partitions define

$$d_c = \sup_s |c_s^b - c_s^a|, \quad d_m = \sup_s \left| \int_0^s (\theta_v^b - \theta_v^a) dv \right|, \quad d_T = \left| \int_0^1 (\theta_v^b - \theta_v^a) dv \right|.$$

The differences are computed from the exact stored controls. No bound on the unknown optimal policy is required.

Let H_{uu}, H_{uc}, H_{cc} bound the absolute utility Hessian entries on $[1.78, 2.42] \times [.7, .8]$, and let A_u, A_c bound its first derivatives on $[1.2, 2.8] \times [.7, .8]$. Put

$$\begin{aligned} P_4 &= \frac{e^{-8}}{2\sqrt{2\pi}}, \quad D_4 = \min\{12, 2(A_u d_m + A_c d_c)\}, \\ \Omega(a, b) &= C_\rho(1) \left[.02(H_{uu}d_m + H_{uc}d_c) + \frac{.01}{Q_r}(H_{uc}d_m + H_{cc}d_c) + D_4 P_4 \right] \\ &\quad + .04e^{-\rho}(.02)d_T + 4\epsilon_{\text{stop}}. \end{aligned} \tag{28}$$

Here ϵ_{stop} is the uniform fixed-policy stopping allowance from the original-payoff evaluator. The same allowance is valid for both policies and all initial states in K .

THEOREM 5—Coupled-payoff initial-state transfer: *Suppose both policies satisfy the checked consumption, preference, zero-portfolio, and wealth-feasibility conditions above. For $z_* = (2, 1.25)$ and $z \in K$,*

$$|\{J_b(z) - J_a(z)\} - \{J_b(z_*) - J_a(z_*)\}| \leq \Omega(a, b). \tag{29}$$

Consequently, a directed central value-difference interval $[d_-, d_+]$ gives the uniform interval $[d_- - \Omega, d_+ + \Omega]$ for the true policy gain on K .

The proof uses a central normal event, a Lipschitz tail allowance, and exact cancellation of the initial-wealth dependence of terminal wealth. The constants are evaluated with directed arithmetic, not fitted to observed state samples. The four corner payoff evaluations used for optimality certification are an additional calculation; they are not used as empirical evidence for the policy-improvement modulus.

9. NEW POLICY-SENSITIVE EVIDENCE IN THE ORIGINAL ECONOMY

Table IV reports every prescribed neural trajectory. At the reference state, the initial bounds are approximately .0392–.0398, and all final bounds are below .01. Unlike the decrease in the old complete-cover certificates, every accepted decrease is accompanied by a disjoint fixed-policy payoff interval in the improving direction. The common optimal upper comparator is consulted only after actor generation; it supplies no training labels.

The coupled-payoff theorem certifies initial-to-final payoff improvement of at least 0.016513 throughout K , for every seed. The four-vertex argument of Proposition 10, now applied to the newly generated neural policies at $k = 2$, gives the uniform bound 0.011816625. The covered

TABLE IV

NEW NEURAL POLICIES: POLICY-SENSITIVE PROGRESS AND OPTIMAL COMPARISON

Seed	B_0	B_{50}	B_{200}	B_{800}	$\inf_K \Delta J$	B_K
19100	0.039437	0.011268	0.009965	0.009764	0.016679	0.011806
19101	0.039587	0.011453	0.009996	0.009766	0.016759	0.011808
19102	0.039755	0.014144	0.010018	0.009775	0.016830	0.011817
19103	0.039190	0.012475	0.010012	0.009757	0.016513	0.011799
19104	0.039502	0.012358	0.009973	0.009765	0.016758	0.011807

B_k is the original-economy regret upper bound at $(0, 2, 1.25)$, adjustment price 2, after k Adam updates. The last two columns are uniform over $K = [1.98, 2.02] \times [1.24, 1.26]$, at time zero. All upper bounds are rounded upward and gain lower bounds downward. These policies are neural-generated time controls with an initial-wealth budget map; they are not the full-state feedback benchmark.

payoff is concave in the initial state, the dual upper comparison is convex, and the quadrature, tail, and stopping allowances are uniform. The numerical check does not infer a continuum conclusion from sampled values.

These two independently verified quantities also give a policy-learning statement. Equation (26) bounds the final-to-initial true-regret ratio by a number below .42 for every seed and every initial state in K . At the reference state, the direct initial-to-final test (25) passes with $\kappa = .75$. Each of the fifteen successive checkpoint transitions also passes that test with $\kappa = .02$. The rates are a posteriori statements about these actual policies. They neither claim convergence of every random initialization nor make a prediction about an unexecuted number of Adam updates.

9.1. A common accuracy frontier and a policy comparison

Table V compares freshly generated SLSQP time transcriptions at 16, 32, and 64 slabs with the neural generators at first attainment of the same reference-state accuracy. The original stopped payoff evaluator and the certified optimal upper comparator are common. The neural method is not faster in this experiment. The comparison establishes actual neural policy improvement and sharp reference-state accuracy, not a speed advantage over direct optimization in a small time-control class.

A further use of Theorem 5 compares each final neural policy directly with the fresh 16-slab SLSQP policy. Uniformly on K , the latter's payoff exceeds the neural payoff by at most 0.000206240. This bound does not suppose that the SLSQP policy is optimal. It supplies a trusted, policy-specific comparison on a nontrivial state set, rather than comparing descriptive action ranges or two loose regret bounds.

The six retained Markov-chain and semi-Lagrangian feedback candidates remain separate objects. Their bounds do not become sharp merely because a time transcription reaches the reference-state target. The supplement reports the positive and negative contributions for all six. At the finest grids the negative policy-side term becomes positive while the positive witness term changes comparatively little. This identifies the component responsible for the reported bound's deterioration; it does not by itself identify whether interpolation, witness fitting, or true policy quality is the ultimate cause.

9.2. Welfare compensation for the neural policies

The new neural certificate has its own budget-feasible economic interpretation. Increase initial wealth by $\delta = .0092$ and add δ/Q_r to every consumption slab of the neural policy. Preference

TABLE V

MATCHED REFERENCE-STATE ACCURACY, WITH A SHARED INDEPENDENTLY CERTIFIED UPPER COMPARATOR

Generator	Updates	Gen. s	Eval. s	Sum s	Bound
19100	200	0.307	8.144	8.451	0.009965
19101	200	0.309	7.916	8.225	0.009996
19102	800	1.215	10.534	11.749	0.009775
19103	800	1.210	10.517	11.727	0.009757
19104	200	0.310	7.892	8.203	0.009973
SLSQP-16	–	0.021	2.635	2.656	0.009722
SLSQP-32	–	0.032	5.266	5.298	0.009704
SLSQP-64	–	0.083	10.495	10.578	0.009699

Neural evaluation cost includes all retained checkpoint checks through first attainment of .01; SLSQP has one final check. Generation includes neural compilation. Costs are conditional on the shared fixed upper comparator. Its historical fitting and verification costs are reported separately in the ledger, not represented as newly executed or silently free. All candidates target the original stopped payoff. SLSQP time transcription is not a Markov-chain or semi-Lagrangian solver.

adjustment and portfolio controls are unchanged. Directed checks over all five policies and all initial wealths in K establish the original consumption cap, identical terminal wealth, and no wealth exit. Intermediate wealth is weakly higher, so the preference-exit time is common and the settlement contribution cannot fall.

PROPOSITION 4—Sufficient neural-policy wealth compensation: *For every reported final neural policy and $(u, x) \in K$, the preceding feasible adjustment satisfies*

$$J_k(0, u, x + \delta; \tilde{\pi}) - J_k(0, u, x; \pi) \geq \frac{\delta}{Q_r} (.8)^{-1.2} C_\rho(1)(1 - p_e) > .0119058,$$

where $k = 2$ and $p_e = 2 \exp\{-.58^2 / (2(.05)^2)\}$. Hence $J_k(0, u, x + \delta; \tilde{\pi}) \geq V_k(0, u, x)$.

The increment is 0.736 percent of reference wealth 1.25. It is sufficient compensation with an explicitly changed feasible consumption policy, not the exact minimum compensating variation of an unchanged policy. It applies to the new neural-generated policies on K , not to the old full-domain feedback certificate or solely to the non-neural price library.

10. CONTINUATION TARGETS AND ATTRIBUTION

A fixed-policy continuation value gives a constructive target even when the optimal value is unknown. Fix one newly trained time-control vector, without the initial-wealth transfer used on K . For $x \in [1.25, 2)$, its deterministic wealth remains above .5, and its wealth drift is strictly negative. The upper face is therefore inaccessible before the preference exit or terminal date. Let q_π denote its true stopped policy value with this fixed control context.

PROPOSITION 5—Policy-induced continuation trace: *The one-sided limit $q_\pi(0, u, 2-)$ exists for $u \in [1.98, 2.02]$. The original-utility Gaussian evaluator, with the same uniform stopping allowance and no wealth transfer, encloses this limit. Its certified excess over $g(0, u, 2)$ is at least 6.708771 uniformly on that preference segment for all five new policies.*

The proof does not redefine the absorbing boundary value. A process initialized exactly on the stopping boundary has immediate payoff g ; the displayed object is the limit from interior

initial wealth. This distinction is precisely why forcing a critic to match settlement on this face is inconsistent. The true q_π has the fixed-policy Feynman–Kac characterization on each time slab. It is not an optimal upper witness, and the numerical execution here certifies its continuation trace rather than a global Bellman residual for a learned approximation to q_π .

A continuation-aware critic update can now impose a verified inequality against a feasible-policy continuation target, rather than merely omitting an equality constraint. For an upper witness w , the appropriate necessary requirement is $w(0, u, 2-) \geq q_\pi(0, u, 2-)$, up to an explicitly allocated comparison error. The target is generated from the new policy and original payoff; no optimal-value labels are supplied. Joint control of this trace and the interior positive Bellman residual remains necessary.

The latter qualification is tested directly. In a predeclared intervention, only the final additive bias of each of five old critics is increased until directed arithmetic proves the earlier feasible-policy trace requirement at $(0, 2, 2)$. All actors remain fixed. Every one of the 1,024 boxes is recomputed. Table VI shows that the resulting total bounds are about 22.7–23.4. A scalar lift removes the particular pointwise obstruction but worsens the global certificate. It is neither selected as an improved witness nor used to claim full-domain accuracy.

TABLE VI
A CONTINUATION-TRACE CONSTRAINT IS NECESSARY, NOT SUFFICIENT

Seed	Bias lift	Trace excess	B^+	B^-
17100	8.69086	6.372265	16.464523	6.295713
17101	9.28629	6.372265	16.469820	6.673984
17102	8.66455	6.372265	16.468794	6.199523
17103	9.30010	6.372265	16.470158	6.888356
17104	8.70903	6.372265	16.453049	6.300897

Only the final critic bias changes; actors and all other weights remain fixed. Each lifted critic satisfies the directed point-trace constraint. The recomputed global bound worsens, so the intervention is not described as a successful global witness solve.

10.1. Attribution in every retained feedback seed

Table VII supplies the complete actor–critic interchange, including the signed policy component, for all ten old feedback seeds. Pairing the initial actor with the final critic reproduces the final certificate in every row. The final negative component is zero; the entire remaining bound is the positive optimal-comparison component. Thus the earlier result establishes witness improvement but does not identify actor improvement. The new fixed-policy comparisons in Section 9 address a different, policy-sensitive question with newly generated actors.

The supplement also reports verified upper-face trace excess, the implied fixed-witness floor, and the positive component at all 30 accessible checkpoints. The matched all-face experiment is retained unchanged. Its width-dependent ranking is evidence about finite-budget performance, not a contradiction of the approximation-space consistency theorem and not evidence for a uniform empirical advantage of the accessibility-aware architecture.

11. STOPPING ACCESSIBILITY AND THE NEURAL APPROXIMATION SPACE

A stopping contract need not be the limit of continuation value at every geometrical boundary point. This distinction matters for a degenerate controlled diffusion: an investor can eliminate

TABLE VII
FULL-DOMAIN ACTOR–CRITIC INTERCHANGE: ALL TEN FROZEN FEEDBACK RUNS

Seed	A_0, C_0	A_1, C_0	A_0, C_1	A_1, C_1	Final B^-
17100	26.300514	23.718686	7.339624	7.339624	0.0
17101	26.499830	24.195287	7.281385	7.281385	0.0
17102	26.326374	23.626229	7.343312	7.343312	0.0
17103	26.511217	24.396419	7.280303	7.280303	0.0
17104	26.367488	23.701299	7.336433	7.336433	0.0
17105	26.468295	24.307436	7.280018	7.280018	0.0
17106	26.352908	23.639651	7.343665	7.343665	0.0
17107	26.666728	24.502208	7.278319	7.278319	0.0
17108	26.339971	23.757239	7.338341	7.338341	0.0
17109	26.527472	24.463132	7.279140	7.279140	0.0

Entries are complete time-zero regret bounds on the original spatial domain, not policy values. The final positive component equals the total in every row; the final negative component is zero. This table audits the earlier full-state experiment and is not relabeled as new training.

portfolio risk and move away from a wealth boundary. It therefore cannot be decided by assigning a larger penalty to every boundary observation in a neural loss.

PROPOSITION 6—Trace consistency in the unchanged economy: *For the economy in Section 2, at $k = 2$,*

$$\liminf_{x \uparrow 2} V_2(0, 2, x) - g(0, 2, 2) \geq 6.45 - .1 \log 2 - 25.1e^{-8} > 6.37. \quad (30)$$

Consequently, a sequence of witnesses continuous on the closed cylinder, each equal to g on its entire stopping boundary, cannot have its uniform positive Bellman residual converge to zero.

The proposition concerns the consistency of an approximation space, not the existence of a solution to the economic decision problem. The original utility, adapted controls, horizon, and liquidation fee are unchanged. Its implication is constructive: enforce the settlement where the proposed policy can stop, and use the correct one-sided comparison where it cannot. In particular, imposing the upper-wealth settlement as a hard equality would exclude the relevant continuation limit. The proof uses the elementary feasible policy $(c, \theta, p) = (.8, 0, 0)$ and does not use an estimated optimal value.

11.1. A signed, facewise comparison

Partition the stopping boundary into measurable faces F_i , assigning intersections to one face. Let

$$\omega_i^*(t, z) \geq \sup_a \mathbb{E}_{t,z} [e^{-\rho(\tau-t)} 1_{F_i}(\tau, Z_\tau)], \quad \omega_i^\pi(t, z) \geq \mathbb{E}_{t,z}^\pi [e^{-\rho(\tau-t)} 1_{F_i}(\tau, Z_\tau)].$$

These are upper bounds on discounted hitting probabilities, not probabilities estimated by simulation. A nonnegative smooth function whose discounted generator is nonpositive and whose stopping trace dominates the face indicator supplies such a bound by optional stopping. A separate invariance argument can give the exact value zero for an inaccessible face.

THEOREM 6—Accessibility-weighted neural verification: *Under Assumption 1, suppose complete enclosures establish*

$$\sup_a \mathcal{R}_a v \leq e_+(s), \quad -\mathcal{R}_\pi v \leq e_-(s), \quad e_+, e_- \geq 0,$$

and $(g - v)_+ \leq \beta_i^+$, $(v - g)_+ \leq \beta_i^-$ on F_i . For an interior initial state,

$$\begin{aligned} 0 \leq V(t, z) - J(t, z; \pi) &\leq \int_t^T e^{-\rho(s-t)} \{e_+(s) + e_-(s)\} ds \\ &+ \sum_i \beta_i^+ \omega_i^*(t, z) + \sum_i \beta_i^- \omega_i^\pi(t, z). \end{aligned} \quad (31)$$

The upper comparison retains the entire original action set. Inaccessibility for the candidate policy does not remove an action from that comparison.

This theorem retains the stopped verification identity underlying Theorem 1. The new distinction is which sign of a trace discrepancy must be paid, and under whose stopping distribution it is paid. Replacing both signs by a common supremum discards precisely the boundary information identified by Proposition 6. The inequality is still an a posteriori certificate; convergence of a candidate-generation procedure is a separate statement.

11.2. An executable accessibility-preserving architecture

Let N_c, N_θ, N_p, N_v be smooth neural outputs. We use

$$\begin{aligned} c &= .425 + .375 \tanh N_c, & \theta &= .2 \tanh N_\theta, \\ p &= \frac{2-x}{1.5} \{ .15 + .65 \tanh N_p \}, \end{aligned} \quad (32)$$

$$v = G + (1-t) \{ -8 + B(u, x) \text{softplus}(N_v) \}, \quad (33)$$

$$B(u, x) = (1 - e^{-12(u-1.2)})(1 - e^{-12(2.8-u)})(1 - e^{-4(x-.5)}).$$

Here $\text{softplus}(y) = \log(1 + e^y)$. Consumption and preference controls obey the original bounds. The portfolio head also obeys the original bounds throughout the state domain. Its boundary factor is part of the policy, not a replacement of the economy's portfolio opportunity.

PROPOSITION 7—Exact elimination of the trace budget: *For an interior initial state, the policy in (32) cannot hit $x = 2$ before another stopping event. The witness in (33) equals g on the two preference faces, the lower wealth face, and the terminal face, and is at least g on the upper wealth face. Hence every trace contribution in (31) is zero.*

The statement remains true for bounded admissible perturbations of the action heads, clipped to their stated intervals *before* multiplication of the portfolio head by $(2-x)/1.5$. An unrestricted additive error in the final portfolio output is different: it can destroy inaccessibility and is not covered by this claim. The replication records this implementation contract explicitly. The certificate describes a real mathematical feedback with stored dyadic weights; it is not a claim about an undocumented floating-point activation routine or an SDE simulator.

The architecture does not by itself make the residual small. Full-domain derivative enclosures must still validate both terms in (31). Reporting a zero trace budget without the remaining

residual budget would therefore be incomplete. Nor do we force the numerical continuation witness to equal the known immediate-settlement value at an initially stopped upper-boundary state: at that state every policy has value g and regret zero directly.

12. AN AUXILIARY FINITE-BELLMAN POLICY-ITERATION BOUND

The following result is an auxiliary statement about a stationary finite Bellman problem, not the algorithm executed in the nonlinear experiments. The executed certificate-aligned algorithm is defined in Section 13. It is useful to distinguish three sources of approximation: policy evaluation, policy improvement, and approximation of a continuous-time transition problem. We state a finite Bellman result that makes the first two requirements quantitative. It does not assume that a particular stochastic-gradient optimizer attains them.

Consider a finite discounted Markov problem with rewards r_a , stochastic or substochastic transition matrices P_a , and contraction modulus $q \in (0, 1)$. Absorbing settlement is incorporated in the reward. Write $T_a v = r_a + qP_a v$ and $Tv = \max_a T_a v$. A neural evaluation v_n of policy π_n and an improved policy π_{n+1} are accepted with recorded defects

$$\|v_n - T_{\pi_n} v_n\|_\infty \leq \epsilon_n, \quad 0 \leq Tv_n - T_{\pi_{n+1}} v_n \leq \eta_n \mathbf{1}. \quad (34)$$

On a continuous state space the same inequalities require a complete enclosure, not a finite collection of collocation residuals. On a finite state space they can be checked by enumeration. They give a stopping and refinement criterion for a learned representation without identifying its training loss with its Bellman defect.

THEOREM 7—A quantitative certified policy-iteration recurrence: *Let $E_n = \|V^* - V^{\pi_n}\|_\infty$ for the finite Bellman problem. If (34) holds, then*

$$E_{n+1} \leq qE_n + \frac{2q\epsilon_n}{(1-q)^2} + \frac{\eta_n}{1-q}. \quad (35)$$

Thus geometrically decreasing certified defects imply convergence of policy values, with the explicit convolution bound obtained by iterating (35). If both defects are bounded by Cr^n , $r \in (0, 1)$, the bound decreases at rate $O(\max\{q, r\}^n)$ when $q \neq r$, and $O(nq^n)$ when $q = r$.

The contraction and policy-improvement ingredients are classical. The operational content is the use of independently checked evaluation and continuous-action defects as acceptance criteria. A neural optimizer that does not achieve those defects has not met the hypothesis. In particular, the full-horizon experiments below are not assigned a convergence guarantee by virtue of being called policy iteration.

A time or state discretization requires an additional comparison. If a numerical optimal value differs from continuous-time value by at most ζ_h , and the numerical value of the transferred policy differs from its continuous-time payoff by at most ξ_h , its continuous-time loss is at most $E_n + \zeta_h + \xi_h$. Neither term is assumed zero. An alternative is to bypass both transfers and certify the actual continuous-time feedback with Theorem 6; that is the procedure used for the classical candidates below. At an inward stopping face, imposing an inconsistent Dirichlet approximation does not supply the required ζ_h . Proposition 6 explains why this qualification is material rather than technical.

For a fixed positive discount rate, $q = e^{-\rho h}$ approaches one as h decreases. The factors in (35) must therefore be included in any accuracy allocation. The theorem is not a dimension-free complexity result. The original nonlinear economy still uses a full rectangular certificate; the structured extension in Section 19 avoids a state tensor cover for a different, explicitly proved reason.

13. COMPLETE-COVER ENVELOPE PROPOSALS

The complete-cover benchmark has three explicit objects: a policy π_θ , a smooth witness v_ϕ , and a complete-cover certificate. Its target is the original continuous-time policy loss, not the optimal value of a discretized training problem. The architecture in Section 11 supplies admissibility and a zero signed trace budget. The following construction addresses the remaining residuals.

Partition time into $0 = t_0 < \dots < t_N = T$ and cover the spatial domain on slab j by M_j closed boxes. Write

$$m_j = \int_{t_j}^{t_{j+1}} e^{-\rho s} ds, \quad b_{ji}^+ \geq \sup_{(s,z) \in Q_{ji}, a} \mathcal{R}_a v_\phi(s, z), \quad b_{ji}^- \geq \sup_{(s,z) \in Q_{ji}} (-\mathcal{R}_{\pi_\theta} v_\phi)(s, z).$$

Each $b_{ji}^\pm$ is nonnegative; the optimal comparison ranges over the original continuous action set. Define

$$B(b) = \sum_j m_j \sum_{\sigma \in \{+, -\}} \max\{0, \max_i b_{ji}^\sigma\}, \quad (36)$$

$$S_\tau(a) = \sum_j m_j \sum_{\sigma \in \{+, -\}} \tau \log \left(1 + \sum_{i=1}^{M_j} \exp(a_{ji}^\sigma / \tau) \right), \quad \tau > 0. \quad (37)$$

The extra unit in each logarithm is the zero logit. It is necessary because only the positive part of each signed residual is charged. All bounds in this section are for time zero; replacing m_j by the appropriate discounted remaining-slab masses gives the corresponding bound at another initial time.

THEOREM 8—Trainable envelope and certified policy loss: *Suppose the hypotheses of Theorem 6 hold, the signed trace budget is zero, and the boxes cover the entire state–time domain. For any finite proposal logits a_{ji}^σ , let*

$$d_j^\sigma \geq \max_i (b_{ji}^\sigma - \max\{a_{ji}^\sigma, 0\})_+, \quad D = \sum_j m_j (d_j^+ + d_j^-).$$

Then every interior initial state satisfies

$$0 \leq V(0, z) - J(0, z; \pi_\theta) \leq B(b) \leq S_\tau(a) + D. \quad (38)$$

If the certified endpoints themselves are used as logits, then

$$B(b) \leq S_\tau(b) \leq B(b) + 2\tau \sum_j m_j \log(M_j + 1). \quad (39)$$

A directed upper enclosure of $S_\tau(a) + D$, computed from the exact stored proposal logits and the certified endpoints, is therefore a sufficient, checkable stopping criterion for the policy-loss target.

The theorem applies to the implemented differentiable interval-envelope loss even when its ordinary floating-point arithmetic has not itself been proved enclosing. The correction D covers any endpoint discrepancy between that proposal graph and the independently evaluated

certificate; it is not assumed to be a rounding error of a particular library routine. In the new audit, the logits are recomputed from the stored neural weights and then interpreted as exact dyadic numbers for the directed soft-maximum calculation. Neither differentiation through an ordinary floating-point interval graph nor a small sampled loss is called a certificate.

The smoothing allowance is explicit. With four time slabs, 256 spatial boxes per slab, $\rho = .04$, $T = 1$, and $\tau = .15$, its upper bound is 1.631868. This is a worst-case allowance, not an observed policy loss. Refining the cover while keeping τ fixed need not remove this allowance. A sufficient joint refinement rule is $\tau \log(1 + \max_j M_j) \rightarrow 0$, together with consistent residual enclosures. The theorem does not assert that a nonconvex optimizer reaches a requested envelope level.

13.1. *The executed algorithm*

The complete-cover neural algorithm uses the following rule. Initialize the actor and critic, form the fixed cover, and minimize $S_\tau(a(\theta, \phi))$ by joint Adam updates of their parameters. At each predeclared checkpoint, freeze and serialize both networks. Independently recompute every signed residual box with directed arithmetic and the original action maximization. Reject an incomplete, nonfinite, or failed enclosure. For a complete object, compare its bound B with the incumbent bound, retain the proposal regardless of the comparison, and replace the incumbent only under a strict decrease. Stop at the declared work budget or when the complete bound meets the stated target. The implementation is `guided_neural.py`; the separate checker is `accessibility_certificate.py`.

The reported original-economy run uses checkpoints 0, 100, 400, temperature .15, learning rate .003, gradient clipping at 100, and two hidden tanh layers of width 16 or 32. There is no point sampling in its envelope objective: all 1,024 boxes enter every proposal update. The ten seeds and paired all-face ablations use matched initialization logic, optimization budget, cover, and arithmetic. The acceptance decision uses the complete MPFR bound, not the smooth proposal loss. The new R18 audit applies Theorem 8 to these already executed, predeclared R17 trajectories; it is not represented as a newly randomized training study.

PROPOSITION 8—What certificate acceptance guarantees: *The incumbent bound of the preceding algorithm is nonincreasing and remains a valid upper bound for its associated incumbent policy. Neither this property nor a decrease in raw proposal bounds implies that the true values of successive policies increase.*

The distinction is substantive. A better witness can certify the same policy more sharply. Conversely, a worse policy can have a smaller bound if it is paired with a much better witness. Accordingly, the numerical record reports raw proposals, incumbent decisions, and actor–critic interchanges separately. A claim of true policy improvement would require an additional policy-value comparison, not just the incumbent rule.

13.2. *A finite-horizon error allocation without a stationary resolvent*

The auxiliary stationary recurrence in Section 12 has a different role. Its amplification factors become large when $q = e^{-\rho h} \uparrow 1$. For a finite horizon, a backward error allocation is more appropriate when a time-discrete implementation is analyzed directly. Let T_n and $T_n^{\pi_n}$ be monotone Bellman operators with Lipschitz modulus q , and let v_N approximate the terminal reward g .

PROPOSITION 9—Finite-horizon evaluation and improvement defects: *If*

$$\|v_N - g\|_\infty \leq b, \quad \|v_n - T_n^{\pi_n} v_{n+1}\|_\infty \leq \epsilon_n, \quad 0 \leq T_n v_{n+1} - T_n^{\pi_n} v_{n+1} \leq \eta_n \mathbf{1},$$

then the nonstationary policy $\pi = (\pi_0, \dots, \pi_{N-1})$ satisfies

$$\|V_0^* - V_0^\pi\|_\infty \leq 2q^N b + \sum_{n=0}^{N-1} q^n (2\epsilon_n + \eta_n). \quad (40)$$

For $q = e^{-\rho h}$, $Nh = T$, and $\epsilon_n \leq \bar{\epsilon} C_\rho(h)$, $\eta_n \leq \bar{\eta} C_\rho(h)$, the accumulated defect term is at most $(2\bar{\epsilon} + \bar{\eta}) C_\rho(T)$.

The finite-horizon statement is representation independent. Its 80 exact-rational finite-MDP regression cases do not constitute execution of a continuous-time neural policy-iteration scheme. The original-economy algorithm instead applies Theorem 8 directly to the continuous-time policy, avoiding an unproved discretization-transfer bound. Both finite-Bellman results are retained, with their different hypotheses and numerical roles made explicit.

14. A SHARP ECONOMIC COMPARISON CONSTRUCTION

A neural certificate should be compared with independently identified bounds for the same economy. We therefore retain the affine-preference market-deflator construction and reimplement its numerical evaluation. The lower policy and the upper comparison object are identified separately from the neural feedback throughout.

14.1. Feasible policies and the original stopping contract

A library policy consists of 16 exact stored consumption constants $c_j \in [.7, .8]$, preference actions $\theta_j \in [0, .2]$, and $p_j = 0$. These controls are held on their time intervals until the original first exit. Wealth is deterministic and satisfies

$$x_1 = e^{.02} \left(x_0 - \int_0^1 e^{-.02s} c_s ds \right).$$

The program proves $x_1 > .5$ and the complete action bounds before evaluating a payoff. It does not round a budget residual to zero or assume that a nearly feasible policy is feasible.

The previous implementation evaluated a polynomial representation with Gaussian moments. The new evaluator instead applies an interval-certified Gauss rule directly to the original CRRA expression on a covered Gaussian domain. After $t = v^2$, it integrates over v and a standard-normal coordinate z . Rationally isolated Gauss nodes, interval weights, and analytic complex-neighborhood remainder bounds enclose the integrals. The unbounded normal tail and the discrepancy from stopping are accounted for analytically. The use of an auxiliary bounded full-horizon extension in this proof does not change the objective (5): it is connected to that objective by an explicit stopping correction, less than 3.87×10^{-18} on K for these policies.

At $k = 4.25$, the second implementation encloses the policy payoff in

$$[-1.315469674845588, -1.315469674834266], \quad (41)$$

with the stored unrounded interval width approximately 1.13×10^{-11} . This is a policy-payoff width, not an optimality gap. The latter still requires an upper bound on V_k .

14.2. The affine-preference dual

Let $dY_s = .02Y_s ds - .3Y_s dZ_s^x$. Its product with wealth satisfies

$$d(Y_s x_s) = (.04Y_s x_s - Y_s c_s) ds + \text{a local martingale.}$$

The cancellation holds for every original adapted portfolio control. Define

$$\Psi(u, y) = \max_{.05 \leq c \leq .8} \left\{ \frac{c^{1-u}}{1-u} - yc \right\}, \quad S_0(u, y) = \Psi(u, y) + .01y - .004 \log(.5), \quad \beta = \partial_u \Psi.$$

For a deterministic reference $\bar{\theta} \in [0, .2]$, set $m(t) = 2 + \int_0^t \bar{\theta}_s ds$ and let Q be the affine tangent to $\phi(u) = -.02(u-2)^2$ at $m(1)$. On $u \in [1.5, 2.5]$, $y \in [.2, 12]$, the supporting inequality is

$$S_0(u, y) \leq S_0(m, y) + \beta(m, y)(u - m), \quad 2 \leq m \leq 2.2. \quad (42)$$

It is justified by analytic monotonicity and complete interval covers, not by sampled tangency or presumed global concavity. Its new primitive checks have strictly positive margins; the supplement gives the proof and the recorded numerical margins.

Put $\hat{y} = \text{proj}_{[.2, 12]} y$ and replace the source by the right side of (42) at $\hat{y}$. The resulting full-line auxiliary problem $A_k(t, u, y)$ retains adapted preference actions, quadratic adjustment cost, and terminal payoff Q . Its source and terminal value are affine in the initial preference. The following positive localization potentials reconnect it to the original first-exit problem:

$$Z_u(h, u) = 8h \{ e^{-120(u-1.5)+42h} + e^{-120(2.5-u)+42h} \}, \quad (43)$$

$$Z_y(h, y) = 8h \{ e^{-22 \log(y/.2)+22.33h} + e^{-22 \log(12/y)+22.33h} \}. \quad (44)$$

For $y_0 \in [1.2, 2.4]$ the resulting upper bound is

$$V_k(0, 2, 1.25) \leq .75y_0 + .1 \log(.5) + A_k(0, 2, y_0) + Z_u(1, 2) + Z_y(1, y_0). \quad (45)$$

The auxiliary value has a computable upper bound consisting of one-Gaussian source expectations, a terminal tangent, a correlated-shock correction, a reference-action gap, and a conditional-variance allowance. The latter is necessary because the adapted marginal value of adjustment is random; replacing it by its unconditional mean without a correction would not be valid. The supplement states and proves the complete formula, including the factors .00375 and .09/(24k) in the covariance and variance terms.

In the second implementation, the source uses separate analytic formulas for its capped and interior consumption branches. Interval automatic differentiation supplies Hessian enclosures, and a rational-root Gauss construction encloses the weighted moments. This differs from the inherited manually differentiated source and polynomial moment integrator. The analytic dual theorem is shared; its mathematical validity is not established merely by agreement between programs. Its finite supporting-plane checks and all decisive numerical quantities are recomputed by the new implementation.

15. JOINT INITIAL-STATE AND PRICE CERTIFICATION

The state-domain extension is obtained without assuming a Lipschitz modulus for the unknown optimal value. Two structural properties suffice: a convex upper comparison function and a concave feasible-policy lower function.

15.1. *Extending the dual in the initial state*

For the fixed dual witness, define

$$b_0 = \int_0^1 e^{-\rho s} \mathbb{E} \beta(m(s), \hat{Y}_s) ds + e^{-\rho} Q'. \quad (46)$$

An interval for b_0 is obtained in the same independent execution as the dual. Since the coefficient of the initial u in the auxiliary criterion is independent of the chosen adapted preference action,

$$A_k(0, u, y_0) = A_k(0, 2, y_0) + (u - 2)b_0.$$

Let U_0 be a certified upper endpoint in (45). It follows that

$$\begin{aligned} V_k(0, u, x) &\leq U_0 + y_0(x - 1.25) + (u - 2)b_0 \\ &\quad + 8e^{-18} \{e^{120(u-2)} + e^{-120(u-2)} - 2\}, \end{aligned} \quad (47)$$

for the verified initial rectangle. Interval multiplication is sign-aware when $u - 2 < 0$. With an interval for b_0 , its upper support function is convex in u . Thus the computable right side of (47) is convex in (u, x) . This is a property of the upper comparison object, not an assertion that V_k itself is convex in wealth.

15.2. *A budget-preserving policy map*

Write $Q_r = \int_0^1 e^{-.02s} ds$. For each stored policy, define

$$c_j(x) = c_j + \frac{x - 1.25}{Q_r}, \quad \theta_j(x) = \theta_j, \quad p_j(x) = 0. \quad (48)$$

Its terminal wealth is identical to that of the central-state policy. All shifted consumption values remain in $[\cdot, \cdot]$ on K ; this is checked for every library member. The policy depends on the initial state and is thereafter a time control until exit. It is not described as an approximate neural feedback surface.

The utility $f(c, u) = c^{1-u}/(1-u)$ is jointly concave over the region needed by the Gaussian cover. With $r = u - 1$ its Hessian satisfies

$$\begin{aligned} f_{cc} &= -uc^{-u-1}, \\ f_{cu} &= -(\log c)c^{-u}, \\ f_{uu} &= -\frac{c^{-r}}{r^3} \{(r \log c + 1)^2 + 1\}, \\ \det D^2 f &= \frac{c^{-2u}}{r^3} \{r^2(\log c)^2 + 2r(r+1)\log c + 2(r+1)\}. \end{aligned}$$

On $u \in [1.2, 2.8]$, $c \in [\cdot, \cdot]$, one has $r(-\log c) < 1$, so the determinant is positive. The covered expected payoff, with the exact budget shift, is therefore concave in the initial state. A single uniform tail and stopping correction turns this covered functional into a lower bound for the original stopped policy throughout K .

PROPOSITION 10—Four vertices and a continuum of prices: *Suppose, at each price node k_i , a convex upper function $U_i(z)$ bounds $V_{k_i}(0, z)$ on a rectangle K . For each policy j , suppose a concave lower function $F_j(z; k)$ bounds its original payoff on K , is affine in k , and has a verified expenditure slope interval. Then corner values alone produce a uniform regret certificate on $K \times [k_1, k_N]$ by the following finite construction. Between adjacent price nodes use the chord of U_i and U_{i+1} . For each policy transfer its four lower corner values affinely in price, using the appropriate expenditure endpoint. At each price select the policy minimizing the largest of its four upper-minus-lower corner gaps. The supremum of the resulting bound is attained at an endpoint or an intersection of the finitely many affine corner-gap functions.*

The optimal value is convex in k because the policy set, dynamics, and stopping contract are common and fixed-policy payoffs have the form $A(\pi) - kB(\pi)$. This justifies the upper price chord, not interpolation of V as though it were affine. The expenditure used for the concave covered lower functional is its full-horizon deterministic expenditure, with the stopping discrepancy already paid uniformly at the largest price. For the separate central-state exact-policy envelope, the program instead bounds the stopped expenditure. Confusing these two slopes would invalidate the transfer; they are separately recomputed and labeled in the evidence.

PROPOSITION 11—Executed state–price certificate: *For the unchanged economy (5), the 18 stored policies with the state map (48) and the price selector in Proposition 10 satisfy*

$$\sup_{(u,x) \in K} \sup_{k \in [.5, 8]} \{V_k(0, u, x) - J_k(0, u, x; \pi_{k,x})\} \leq 0.012095431554. \quad (49)$$

The central-state selector based on the independently evaluated stopped policies satisfies

$$\sup_{k \in [.5, 8]} \{V_k(0, 2, 1.25) - J_k(0, 2, 1.25; \pi_k)\} \leq 0.009969007711. \quad (50)$$

The execution evaluates all 72 policy–state corners, uses all 18 independently verified upper witnesses, and checks 211 exact candidate breakpoints for the joint state–price construction. No unsampled state requires an empirical smoothness extrapolation. The exact rational worst-case joint bound is

$$\frac{2187960974729572475442514300375}{180891518006739738096917284913152}.$$

The displayed decimal in (49) is rounded upward. The computational record contains the complete selected-policy regions and every corner interval. Appendix C proves the interpolation and finite-breakpoint steps.

16. CERTIFICATE-ALIGNED COMPUTATION ON THE ORIGINAL ECONOMY

All experiments in this section use the original economy, its full continuous action comparison, $k = 2$, and the complete state–time domain. They are not price-library or inventory results. The immutable R17 scientific object contains the prescribed trajectories and all raw certificates. The new R18 calculations audit their objective implications, reproduce two full-domain certificates, and reconstruct every displayed aggregate from the retained records.

TABLE VIII
ALL PRESCRIBED ORIGINAL-ECONOMY NEURAL TRAJECTORIES

Seed	Width	B_0	B_{100}	B_{400}	All-face B_{400}	Seconds
17100	16	26.300514	10.104267	7.339624	7.345561	104.1
17101	32	26.499830	7.547734	7.281385	7.263538	193.4
17102	16	26.326374	10.237435	7.343312	7.351242	96.1
17103	32	26.511217	7.535185	7.280303	7.263378	143.4
17104	16	26.367488	10.591643	7.336433	7.345015	71.3
17105	32	26.468295	7.555938	7.280018	7.261518	114.4
17106	16	26.352908	9.701133	7.343665	7.349999	51.2
17107	32	26.666728	7.604406	7.278319	7.262408	97.2
17108	16	26.339971	9.745279	7.338341	7.345879	48.6
17109	32	26.527472	7.553444	7.279140	7.261511	79.0

All entries use the full original state–time cylinder and continuous action upper comparison. B_j denotes a raw proposal certificate, not an incumbent replacement. Seconds include all three certificate checks and the complete accessible training trajectory. Both architectures use identical declared budgets. Printed values are rounded; exact directed bounds, signed decompositions, and all 60 objects are retained.

16.1. Raw trajectories, acceptance, and the training objective

Table VIII reports every seed. All ten accessibility-aware raw sequences decrease strictly at both transitions; all paired all-face sequences also decrease. Thus the improvement is not created by selecting the incumbent or hiding a rejected proposal. All 40 post-initialization proposals are accepted by the strict rule. The historical sampled-training experiment is different: all ten of its 800-to-2,400-update certificates deteriorate, although sampled residual mean squared error falls in nine of ten cases. The new training loss reverses that particular anti-alignment in the observed experiment; it is not a proof that every longer Adam run must improve the certificate.

The complete-cover objective audit covers 60 neural objects and 61,440 cells. It verifies all 589 files in the immutable scientific manifest, checks network/certificate identities and complete cell coverage, and re-evaluates the proposal logits. Directed evaluation of the soft maximum and its endpoint correction verifies Theorem 8 for every object. The largest integrated correction D is below 1.20×10^{-13} ; the remaining differences between a smooth objective and a box maximum are not an arithmetic precision effect. The audit is deliberately separate from the training experiment and retains the exact recomputed logits.

The final accessibility-aware bounds are between 7.2783190 and 7.3436651, and none reaches .01. The original target is not relaxed. The result establishes a reproducible improvement in the certified objective, not sharp accuracy of the original nonlinear neural policy. Training, verification, and checkpoint selection costs are included separately in the retained resource records. At the final checkpoint, the observed mean total trajectory cost is 99.87 seconds; timings describe the recorded hardware and concurrent workload, not a matched-accuracy speed ranking.

16.2. A matched boundary ablation and a fixed-witness floor

At width 16, the accessibility-aware final bound is smaller than its paired all-face bound in all five pairs. At width 32, the ranking reverses in all five pairs. The paired mean difference (accessibility-aware minus all-face) is approximately .005049. This is not evidence of a uniform finite-budget advantage for the corrected trace class. The structural result in Proposition 6 remains an asymptotic consistency statement, while the actual finite-budget comparison depends on the learned witness and optimization trajectory.

For seed 17100, refining a fixed final accessibility-aware object from 1,024 to 8,192 and 65,536 cells changes its bound from 7.339625 to 7.280222 and 7.251560. The corresponding all-face bounds are 7.345561, 7.267777, and 7.236151. These are verifier-refinement curves around fixed objects, not new policy training. They quantify enclosure dependence without suggesting that a sufficiently fine cover would reach .01.

PROPOSITION 12—A computable fixed-witness obstruction: *Let v satisfy the upper-comparison trace conditions and let $h_v = v(0, 2, 2) - g(0, 2, 2)$ be its upper-wealth trace excess. Write $\Delta = 6.45 - .1 \log 2 - 25.1e^{-8}$. Any nonnegative, time-dependent bound on its positive optimal residual must satisfy*

$$\int_0^1 e^{-.04s} e_+(s) ds \geq (\Delta - h_v)_+. \quad (51)$$

This is a lower bound on the certificate component of this fixed witness, not a lower bound on the regret of its policy.

The final accessible critics have verified trace excesses only about .052–.097. Their lower floors from (51) range from 6.274882 to 6.320490. Consequently, refining their covers or increasing arithmetic precision cannot close the original accuracy gap. A materially different continuation witness is required. The all-face class has the stronger structural obstruction $h_v = 0$.

The actor–critic interchange sharpens this diagnosis. Pairing the initial actor with the final critic for seed 17100 gives exactly the same retained bound, 7.339624178645005, as the final actor/final critic pair. Pairing the final actor with the initial critic gives 23.718685694842687. Thus the certificate descent cannot be attributed to a demonstrated increase in the actor’s true payoff. Both facts are retained rather than reinterpreted as policy improvement.

16.3. Unrestricted candidate generation and independent witnesses

The new controlled Markov-chain and semi-Lagrangian candidates do not multiply their interior portfolio choices by the neural wealth-distance factor. Each solves over the original bounded action grid, including zero portfolio. The upper-wealth grid column represents the interior continuation trace: zero-portfolio continuation is computed and compared with immediate settlement for nonzero portfolio at that trace. The retained policies select zero portfolio on this column. The certified controls are the original-action-clipped, exact-real bilinear state interpolants, constant within each time interval. Their interpolation consequently has bounded $p/(2 - x)$ and cannot hit the upper face before another stopping event; no interior distance multiplier is imposed.

Each candidate receives a separately trained witness and a full continuous-time residual check against the original continuous action set. Hence neither the discrete central value nor a common inaccurate neural witness is used as its accuracy certificate. Table IX gives all six resulting bounds and separates candidate generation, witness construction, and verification. The method-specific witnesses remove the earlier common-witness confound but do not produce a sharp matched-accuracy frontier. In particular, the finest-grid bounds increase because their policy residual contribution is no longer zero. The numerical values of the discrete schemes remain diagnostics, not certified continuous-time values.

16.4. Control changes and numerical scope

The supplement also reports whole-domain ranges for all six classical policies and whole-price/time ranges for all twenty fresh-library policies. For the latter, consumption lies in

TABLE IX
ORIGINAL-ACTION CLASSICAL CANDIDATES WITH METHOD-SPECIFIC WITNESSES

Method	Nodes/axis	Time steps	Bound	Generate	Witness	Verify
MC	9	64	7.339815	0.34	62.14	9.41
MC	17	128	7.314510	7.60	62.41	9.71
MC	25	256	7.763463	124.01	39.84	6.71
SL	9	64	7.342250	0.35	67.46	8.60
SL	17	128	7.314559	8.52	45.80	7.58
SL	25	256	7.774123	74.71	39.22	5.34

Times are seconds. MC: controlled Markov chain; SL: semi-Lagrangian. Interior actions use no neural distance multiplier. Every transferred policy has a separate witness and a complete continuous-time check; all six fail the unchanged .01 target. These bounds do not rank true policy values. Discrete central values remain diagnostics.

[.7618225, .7721747], preference adjustment in [.0031022, .2], and portfolio is exactly zero. These library ranges hold for every policy selected over $k \in [.5, 8]$, not only the displayed price nodes.

Complete-box bounds on the original neural controls are reported in the supplement for all ten final actors. For seed 17100, the final ranges are approximately [.530998, .749371] for consumption, [.093966, .113275] for preference adjustment, and [0, .087417] for portfolio. The corresponding uniform bounds on the change from initialization are .218738, .019170, and .100661. These are conservative ranges of mathematical control functions over the full state–time domain, not simulated confidence intervals or policy-optimality claims.

The certificate interprets stored weights as exact dyadic numbers and activations as mathematical tanh/softplus. The allowed head perturbation is 2^{-24} and precedes the wealth-distance factor. Neither the proposal graph’s ordinary floating-point activations, a deployed inference engine, nor an SDE simulator is certified by this contract. Independent arithmetic reduces rounding risk but does not independently establish every shared analytical premise.

17. NONLINEAR MULTI-PRODUCT CONTROL WITH A STATE-UNIFORM CERTIFICATE

A high-dimensional test should require the policy to depend on the high-dimensional state. We therefore distinguish the following nonlinear planning problem from the two-mode quadratic inventory model retained in Section 19. Here there is no Riccati reduction or supplied optimal solution.

There are d heterogeneous products and $H = 12$ dates. Initial inventory is $x \in [-1, 1]^d$. The deterministic dynamics and total cost are

$$z_0 = x, \quad z_{h+1} = \frac{3}{5}z_h + u_h - b_h, \quad b_{hi} = \frac{((h+i) \bmod 7) - 3}{100}, \quad (52)$$

$$F_x(U) = \sum_{h=0}^{H-1} \{2\|u_h\|_2^2 + \Phi(z_{h+1})\}, \quad U = (u_0, \dots, u_{H-1}) \in \mathbb{R}^{Hd}, \quad (53)$$

$$\Phi(z) = \sum_{i=0}^{d-1} \left\{ \frac{q_i z_i^2}{2} + \frac{1}{2} \log \cosh z_i \right\} + \frac{3}{10} \sum_{i=0}^{d-1} \log \cosh(z_i - z_{i+1}), \quad q_i = 1 + \frac{i}{5(d-1)}, \quad (54)$$

where indices in the coupling term are cyclic. Production/adjustment costs are quadratic, while individual and cross-product inventory penalties are nonquadratic. Every coordinate enters

the nonlinear coupling; heterogeneity prevents the two-invariant-mode decomposition of the quadratic comparison model. Its Bellman equation is

$$W_h(z) = \inf_{u \in \mathbb{R}^d} \{2\|u\|_2^2 + \Phi(\tfrac{3}{5}z + u - b_h) + W_{h+1}(\tfrac{3}{5}z + u - b_h)\}, \quad W_H = 0.$$

The computation instead uses the equivalent deterministic control-plan formulation, without treating either formulation's unknown value as training data.

A two-hidden-layer neural map takes all d coordinates of x and produces all Hd controls:

$$U_0(x) = \tfrac{1}{4} \tanh N_\theta(x).$$

It is trained on the original cost, not optimal labels, using 400 Adam updates, batches of 32 uniformly drawn initial states, learning rate .003, width $\min\{64, 2d\}$, and three predeclared seeds. A correction stage applies

$$U_{k+1}(x) = U_k(x) - \frac{16}{209} \nabla_U F_x(U_k(x)). \quad (55)$$

The complete solver is the neural initialization together with the declared number of corrections. The correction, not a change of model or an optimal-value label, supplies its quantitative convergence mechanism.

THEOREM 9—Uniform nonlinear control accuracy and work: *For every $d \geq 2$ and every x , the cost in (53) satisfies*

$$4I \preceq \nabla_U^2 F_x(U) \preceq \frac{177}{8} I, \quad U \in \mathbb{R}^{Hd}. \quad (56)$$

Let $r = 145/209$. For every $x \in [-1, 1]^d$ and every bounded neural initialization above, the exact-real correction obeys

$$0 \leq F_x(U_k(x)) - \inf_U F_x(U) \leq \frac{\|\nabla F_x(U_k(x))\|_2^2}{8} \leq \frac{Hd(8.85)^2}{8} r^{2k}. \quad (57)$$

A gradient evaluation requires $O(Hd)$ arithmetic operations using forward state propagation and backward adjoints. Thus, after initialization, a sufficient number of correction steps for total loss at most ε is

$$k \geq \max \left\{ 0, \left\lceil \frac{\log(Hd(8.85)^2/(8\varepsilon))}{2 \log(209/145)} \right\rceil \right\}.$$

No tensor cover of the initial-state cube is required.

The work statement counts exact-real arithmetic operations, not bit complexity or the cost of floating-point certification. The result uses strong convexity and the sparse coupled dynamics of this application. It is not a theorem for a general stochastic high-dimensional HJB equation. The factor d in total loss is visible, and no per-coordinate normalization is used to hide it.

17.1. Uniform and pointwise numerical objects

Table X reports the directed evaluation of the state-uniform bound for every specified dimension and correction budget. The $d = 128$ total bound is below .000360 after 24 corrections

TABLE X
STATE-UNIFORM TOTAL-COST LOSS FOR NONLINEAR CONTROL

Corrections	$d = 8$	$d = 32$	$d = 128$
0	939.87	3759.48	15037.9
8	2.70779	10.8312	43.3247
16	0.00780122	0.0312049	0.12482
24	2.24755×10^{-5}	8.99022×10^{-5}	0.000359609
32	6.47527×10^{-8}	2.59011×10^{-7}	1.03604×10^{-6}

The domain is every initial $x \in [-1, 1]^d$. Bounds concern the exact-real neural initialization and rational-step gradient correction, not an uncertified floating-point inference graph. Entries are total cost losses, not per-coordinate losses. The theorem holds for every bounded initialization, including an untrained one.

TABLE XI
MATCHED CERTIFICATE COMPARISONS AT 128 COORDINATES

Method	Largest query bound	Train	Online	Verify
Neural + gradient	1.63×10^{-14}	12.709	0.078	0.678
Zero + gradient	2.06×10^{-14}	0.000	0.078	0.555
Accelerated gradient	2.6×10^{-16}	0.000	0.071	0.648
L-BFGS	3.13×10^{-14}	0.000	0.096	0.699

All query plans use the same MPFR KKT-gradient certificate. Gradient methods use 32 corrections; L-BFGS uses its retained stopping result. Neural entries summarize six query/seed plans; classical entries summarize two queries. Times are descriptive means in seconds; neural training is a one-time per-network cost, not a per-query charge. Query bounds do not replace the state-uniform bound in Table X.

and below 1.037×10^{-6} after 32. These inequalities hold for the exact-real neural map and correction at every initial state in the cube, regardless of which of the three stored neural seeds is used.

Separately, the actual stored floating-point plans are checked by MPFR at two fixed initial vectors: an alternating corner and a predeclared pseudorandom vector. The checker evaluates the original cost and KKT gradient using exact rational model constants, and applies the first inequality in (57). It does not assume that the floating-point optimizer has converged. All 90 neural query/budget objects and 66 classical query/budget objects are retained across dimensions. At $d = 128$ and 32 corrections, the six retained neural query-plan bounds range from 8.28771×10^{-15} to 1.62604×10^{-14} . Those sharper values concern those six exact stored plans, not every state in the cube. The new R18 execution rechecks all 18 final neural query plans across the three dimensions.

17.2. Matched certificate comparisons

The comparison methods are zero-start gradient descent with the same step, strongly-convex accelerated gradient, and L-BFGS. All use the same original model and the same independent gradient-based cost-loss inequality. Table XI reports the final $d = 128$ query results and separates neural training, online solution, and verification time. The supplement includes the corresponding tables at $d = 8$ and 32 and every retained correction budget.

The data do not establish a neural speed advantage: strong classical methods attain comparable or smaller certified errors without the training cost. The state-uniform theorem likewise holds for any initialization in the specified bounded class, including an untrained one. Neural initialization

and its amortized cost are therefore reported as components of the solver, not credited with the contraction supplied by the correction. This application establishes an accurately certified, full-state-dependent nonlinear control-plan solver; it does not replace the original stochastic economy or close that economy’s remaining neural accuracy gap.

18. LEARNED QUALITY AND MATCHED ONLINE WORK

The nonlinear inventory problem has strong convexity $m = 4$, Hessian upper bound $L = 177/8$, step $2/(m + L) = 16/209$, and gradient contraction factor $r = 145/209$. The existing state-uniform theorem holds for every bounded initialization, including an untrained one. Training is therefore irrelevant to that particular guarantee. To identify a contribution from learning, the initial approximation quality and the required online work must be measured separately.

THEOREM 10—Learned-gradient correction bound: *For any exact initial plan $f_\theta(x)$, let $G_\theta(x) \geq \|\nabla F_x(f_\theta(x))\|_2$. After j exact-real gradient corrections,*

$$F_x(U_j) - F_x^* \leq \frac{G_\theta(x)^2}{8} \left(\frac{145}{209} \right)^{2j}. \quad (58)$$

On a finite query set, the largest independently enclosed initial gradient gives a uniform bound on that set. More generally, suppose a set S is covered by radius- h Euclidean balls around verified queries and f_θ is K_θ -Lipschitz. Then

$$\sup_{x \in S} \|\nabla F_x(f_\theta(x))\|_2 \leq \max_i G_\theta(x_i) + \left(\frac{177}{8} K_\theta + \frac{87}{16} \right) h$$

holds, and its right-hand side is a valid uniform choice for G_θ in (58).

The last term follows from the state-to-control-gradient cross derivative of this particular model. A directed product of matrix-norm bounds gives a valid K_θ for the stored tanh network. It may be conservative. The theorem does not claim that 32 queries cover a 128-dimensional cube, and the existing initialization-agnostic cube bound is not replaced by an unverified learned-quality estimate.

18.1. A new, predeclared finite query distribution

For each dimension $d = 8, 32, 128$, 32 uniform-cube draws are fixed with seed $19200 + d$ before their evaluation. All three stored neural initializers are compared with zero-start gradient descent and zero-start acceleration at target total losses $10^{-2}, 10^{-3}, 10^{-4}, 10^{-6}$. Each delivered floating-point plan is serialized and independently checked by its directed gradient certificate. Ordinary floating-point stopping diagnostics can request a check but cannot accept a plan. The maximum budget is 64 corrections, and no failed query is removed.

For a fair work count, acceleration returns its checked search point. Thus it does not pay an artificial second gradient call merely because its internal extrapolated and unextrapolated points differ. Every method pays for one gradient per correction and one final gradient evaluation. Neural inference, corrections, final directed checking, and the recorded offline training cost are identified separately.

The learned map reduces online gradient work substantially at 8 and 32 coordinates. At 128 coordinates and tolerance .01, every neural seed uses five mean corrections, compared with six

TABLE XII
MATCHED DIRECTED TOLERANCE ON A NEW FINITE QUERY DISTRIBUTION

d	Target	Neural GD	Zero GD	Accelerated	Wins / 96
8	10^{-2}	1.05	3.44	3.84	96
8	10^{-3}	2.35	5.19	5.12	96
8	10^{-4}	4.27	7.16	6.97	96
8	10^{-6}	8.56	11.56	10.84	96
32	10^{-2}	1.86	4.72	4.94	96
32	10^{-3}	3.31	6.47	6.06	96
32	10^{-4}	5.36	8.66	8.00	96
32	10^{-6}	9.84	13.00	12.00	96
128	10^{-2}	5.00	6.00	5.97	93
128	10^{-3}	7.00	8.00	7.00	0
128	10^{-4}	9.06	10.00	9.00	0
128	10^{-6}	14.00	14.00	13.00	0

Mean correction counts; each method uses one gradient evaluation per correction and one final gradient evaluation. Neural means pool three frozen initializers and 32 predeclared queries. Classical means use the same 32 queries. A win means strictly fewer gradient evaluations than acceleration for the paired query; it is not a wall-time claim. Acceleration returns its independently checked search point. The set of 32 queries is not a cover of the state cube.

for zero-start gradient descent and approximately 5.97 for acceleration; it uses fewer gradients than acceleration on 31 of 32 paired queries per seed. At 10^{-3} the neural and accelerated counts coincide, and at 10^{-6} acceleration needs fewer corrections. These outcomes identify a limited, tolerance-dependent contribution from learning rather than an initialization-independent guarantee misattributed to training.

The finite-query initial-gradient theorem is evaluated as a second, separate work statement. The supplement reports the proved uniform correction budgets on each fixed query set. They can be more conservative than the actual checked stopping counts. Stored floating-point plans receive their own certificates; exact-real contraction is not used as an undocumented bound on a floating-point deployment graph.

18.2. Amortization and the cost of completeness

For a declared query distribution and target, let T_{train} be offline cost and let $\bar{t}_0, \bar{t}_\theta$ be measured mean per-query times. When $\bar{t}_0 > \bar{t}_\theta$, the accounting break-even count is

$$Q_{\text{break}} = \left\lceil \frac{T_{\text{train}}}{\bar{t}_0 - \bar{t}_\theta} \right\rceil + 1.$$

No finite break-even is reported when the measured saving is nonpositive. The supplement gives both solver-only and final-check-inclusive counts, all three seeds, and every target. The training time comes from the stored training execution; query times are five-repeat medians in the new execution. These are explicit transferred-cost estimates, not a matched-hardware retraining experiment, statistical confidence intervals, or certified speedups. In particular, no end-to-end acceleration advantage is asserted for the 128-dimensional experiment.

The complete-cover feedback algorithm has a different scaling object. With M boxes, state dimension d , depth ℓ , and hidden width w , storing the value, time derivative, state gradient, and state Hessian requires $s_d = 2 + d + d(d+1)/2$ components. A direct dense-jet implementation

costs on the order of

$$M\{s_d(dw + \ell w^2) + C_A\}$$

scalar operations, where C_A is the per-box cost of certified action maximization. Direct back-propagation stores on the order of $M\ell w s_d$ intermediate quantities. Streaming verification can replace M by its chunk size in working memory, but does not remove the total box work. For a tensor cover, $M = N_t \prod_i N_i$; interval dependency and action subdivision can increase the required counts. These costs are not suppressed in a dimension-free complexity claim. In contrast, one correction in the sparse nonlinear inventory model costs $O(Hd)$ after neural inference.

19. RETAINED QUADRATIC INVENTORY VALIDATION

For historical validation of the dynamic certificate, retain the quadratic inventory-adjustment economy below. Its numerical gain problem reduces to two scalar Riccati equations; the genuinely vector-state nonlinear experiment is Section 17. With $M = \mathbf{1}\mathbf{1}'/d$,

$$dX_s = (AX_s + a_s)ds + .1dW_s, \quad A = -.2I + .3M, \quad (59)$$

and minimize

$$\mathcal{J}(t, x; a) = \mathbb{E} \left[\int_t^1 \{X'_s(I + .5M)X_s + |a_s|^2\} ds + .5|X_1|^2 \right]. \quad (60)$$

Controls are square-integrable adapted adjustments. Mean inventory affects drift and aggregate holding cost; idiosyncratic inventory also has its own adjustment dynamics and shocks. This is a coupled d -state dynamic problem, not an action maximization at frozen derivatives. It is a structured, uncalibrated application, and the structure is available to classical solvers as well.

The orthogonal and mean modes have $(a_\perp, q_\perp) = (-.2, 1)$ and $(a_m, q_m) = (.1, 1.5)$. Their Riccati gains satisfy

$$(p_i^*)' + 2a_i p_i^* - (p_i^*)^2 + q_i = 0, \quad p_i^*(1) = .5.$$

We train two neural gain outputs $\hat{p}_i(t) = .5 + (1 - t) \text{softplus}(N_i(t))$ against these *differential residuals*, not against exact Riccati solutions. The feedback is

$$\hat{a}(t, x) = -\hat{P}(t)x, \quad \hat{P} = \hat{p}_\perp(I - M) + \hat{p}_m M.$$

The network changes its gain with time and controls all d state coordinates. One learned pair is reused across the six dimensions; this reuse is not counted as six independent training runs.

THEOREM 11—State-global quadratic neural policy certificate: *Suppose interval differentiation on a complete time cover bounds the two neural Riccati residuals by ε_i . Put $\kappa_\perp = 1.4$, $\kappa_m = .8$, and*

$$D = \max_i \varepsilon_i \frac{1 - e^{-\kappa_i}}{\kappa_i}, \quad C = \frac{1 - e^{-.8}}{.8}.$$

For every $d \geq 1$ and every $|x|^2/d \leq 1$,

$$0 \leq \frac{\mathcal{J}(0, x; \hat{a}) - \mathcal{V}(0, x)}{d} \leq D^2 \left\{ C + \frac{.01}{.8} (1 - C) \right\}. \quad (61)$$

The bound concerns the continuous-time controlled diffusion on all of $\mathbb{R}^d$, not a finite collection of initial states.

The proof uses a quadratic performance-difference identity and a closed-loop Lyapunov moment bound. It replaces state covering by algebra valid on the entire state space. The certificate’s time-enclosure work depends on network size and the number of time cells, rather than a tensor grid in d . Applying the feedback requires a mean and coordinatewise operations, hence $O(d)$ work. This advantage comes from the model’s two invariant modes; it is not a generic escape from dimensional dependence for nonlinear Bellman equations.

Ten fixed seeds, two widths, and both predeclared checkpoints are retained. Every final checkpoint has per-coordinate certified loss at most 0.000095606; the complete range and costs appear in Table XIII. The certificate scales to total loss by multiplication by d . The supplement also reports the earlier checkpoint and fixed-network time-cover refinements separately. A structure-aware classical calculation solves only two scalar Riccati equations. Its low cost is reported rather than hidden: this application establishes accurate, reproducibly generated neural feedback with full dynamic certification, not superiority over the classical solution of this specially structured problem.

TABLE XIII
CERTIFIED NEURAL POLICY LOSS PER INVENTORY COORDINATE

Seed	Width	250 updates	1,000 updates	Train (s)	Verify (s)
16200	8	0.00117290	0.00009561	5.28	1.18
16201	16	0.00045470	0.00007661	7.05	2.85
16202	8	0.00051798	0.00006084	5.40	1.27
16203	16	0.00011929	0.00003574	7.44	3.26
16204	8	0.00179439	0.00008973	5.09	1.09
16205	16	0.00054105	0.00002530	6.28	3.05
16206	8	0.00036825	0.00004136	5.59	1.36
16207	16	0.00023854	0.00009053	6.34	3.34
16208	8	0.00099196	0.00005105	5.88	1.60
16209	16	0.00036174	0.00007427	5.08	2.29

Each checkpoint uses a complete 512-cell time cover and a state-global quadratic proof. Bounds hold for every $|x_0|^2/d \leq 1$. Multiply the reported bound by d for total loss at $d = 4, 8, 16, 32, 64, 128$. One gain pair is reused across dimensions. Timings cover one training trajectory and its two checks, not six duplicated runs.

19.1. A rigorously validated classical comparison for the quadratic model

The preceding quadratic model has only two independent gain equations. We retain it as a transparent validation case, not as the nonlinear high-dimensional benchmark. Its classical comparison is now certified rather than merely an RK45 diagnostic.

For each scalar mode, with coefficients $(a, q) = (-.2, 1)$ or $(.1, 1.5)$ and terminal gain $p(0) = .5$ in remaining time h , let $\kappa = \sqrt{a^2 + q}$ and $y = (.5 - a)/\kappa$. The exact gain is

$$p^*(h) = a + \kappa \frac{y + \tanh(\kappa h)}{1 + y \tanh(\kappa h)}.$$

It is increasing on $[0, 1]$. Directed endpoint values therefore enclose its full range on each time cell. The implemented policy stores a dyadic midpoint gain on each cell; MPFR bounds its

TABLE XIV
VALIDATED CLASSICAL POLICIES FOR THE TWO-MODE QUADRATIC MODEL

Time cells	Per-coordinate bound	Total bound, $d = 128$	Seconds
32	0.000304119	0.0389272	0.064
128	1.91901×10^{-5}	0.00245634	0.127
512	1.20221×10^{-6}	0.000153883	0.380
2048	7.51824×10^{-8}	9.62334×10^{-6}	0.929

Complete time-cell range enclosures use the closed-form scalar Riccati solutions and MPFR-directed arithmetic. The policies are dyadic piecewise-constant gains. Both generation and validation are included in seconds. This is the structured quadratic comparison, not the nonlinear application.

uniform gain error δ . All stored gains are strictly above .5, so the moment estimate used in the quadratic performance-difference identity gives per-coordinate loss at most

$$\delta^2 \left\{ C_{.s}(1) + \frac{.01}{.8} (1 - C_{.s}(1)) \right\}.$$

Total loss is this quantity multiplied by d . Table XIV reports both units and includes the full generation-and-verification cost. In particular, the 128-cell classical policy has a $d = 128$ total bound below .002457, and the 2,048-cell policy has a total bound below 9.624×10^{-6} . This is the appropriate rigorous comparison for a problem whose exact structure is so favorable to classical methods.

20. HISTORICAL SAMPLED-TRAINING AND RESTRICTED-BASELINE EXPERIMENTS

This section retains the R16 experiment without substituting later results for its observations. The original nonlinear economy is unchanged in all experiments in this section. The architecture is (32)–(33). Seeds 16100–16109 use two hidden tanh layers, with width 16 for even seeds and width 32 for odd seeds. Every trajectory retains the 800- and 2,400-update checkpoints. Each actor update follows three critic updates; Adam learning rates are .0002 and .0005, respectively, with batch size 128, double precision, and one Torch thread. A quarter of each sampling mixture targets preference and lower-wealth boundary layers. No optimal-value labels are used. These settings were fixed before the canonical remote execution; no best-seed substitution is made.

For every checkpoint, MPFR interval differentiation encloses the complete $[0, 1] \times [1.2, 2.8] \times [.5, 2]$ cylinder on $4 \times 16 \times 16$ cells. The continuous consumption, adjustment, and portfolio maxima use the original action set. The certificate allows a componentwise head perturbation of 2^{-24} , clipped to the head bounds before the portfolio-distance multiplication. This location of the perturbation matters: an unrestricted additive error in the final portfolio could invalidate the boundary-inaccessibility argument. The proof concerns mathematical real-valued networks with stored dyadic weights under this explicit output contract. It does not certify a particular machine-library tanh implementation or an SDE simulation stack.

Table XV retains every prescribed checkpoint. The exact trace budget is zero, but the positive optimal residual and negative policy residual still produce large bounds. An increase in optimization budget uniformly loosens these certificates. Since the bounds also include interval dependency, this observation does not by itself prove that the realized policy payoff deteriorates. The refinement experiment holds the first seed’s final network fixed and doubles the cover in each coordinate; the complete cell records separately retain the signed budget and the more conservative unsigned residual/action budget. The hard-all-face ablation is retained as well.

TABLE XV
FULL-HORIZON NEURAL CHECKPOINTS IN THE ORIGINAL ECONOMY

Seed	Width	800 updates	2,400 updates	Train (s)	Verify (s)
16100	16	30.19468	54.00405	165.70	28.36
16101	32	41.45535	96.87386	180.11	79.46
16102	16	28.84397	53.60464	156.26	29.22
16103	32	42.60784	93.28879	162.35	59.47
16104	16	29.69505	68.44466	138.88	23.06
16105	32	48.37134	100.68512	139.71	65.74
16106	16	30.03684	59.19474	129.16	27.60
16107	32	45.64637	102.35557	143.36	62.72
16108	16	28.54732	49.67500	132.95	25.88
16109	32	44.11682	95.77234	74.49	23.67

Each bound is uniform over the original interior state rectangle at time zero, $k = 2$, and uses 1,024 cells with the original continuous action maximum. The trace contribution is exactly zero. Training time covers both checkpoints in one trajectory; verification time sums their two audits. No checkpoint meets 10^{-2} . Resource measurements are descriptive, not matched-accuracy efficiency comparisons.

Its finite-resolution certificate is not evidence that the excluded approximation space becomes consistent as residual error tends to zero.

20.1. State-space candidate generators

We implement a controlled Markov-chain approximation with nonnegative transition weights and a semi-Lagrangian weak Euler approximation with correlated shocks. Each uses the same utility, discount, action bounds, state domain, and first-exit settlement. Refinement uses 9, 17, and 25 nodes per state coordinate, 64, 128, and 256 time steps, and progressively richer bounded control-head grids. The Markov-chain implementation uses wide diffusion jumps and a checked nonnegative stay probability; the semi-Lagrangian method uses four symmetric shock directions. Approximate trajectories apply the liquidation contract at their first straight-segment exit. This discretization is a candidate-generation device, not an asserted exact solution of the stopped diffusion.

For verification, each numerical policy is extended by bilinear interpolation in state and piecewise-constant interpolation in time. Its portfolio head is multiplied by $(2 - x)/1.5$, exactly as in the neural candidate class. These are feasible policies in the unchanged economy, although the candidate class is narrower than the original collection of all admissible controls. The upper comparison still uses every original control. We enclose the interpolated policies with the same complete verifier and the fixed seed 16100 final critic. Thus their discrete value outputs are never used as continuous-time upper or lower bounds.

The common witness standardizes the verification contract but is inaccurate; these loose bounds are not a comparison of solver accuracy. None of these six candidates attains the 10^{-2} target under that certificate. The experiment supplies two distinct state-space constructions, three refinements, and actual continuous-time policy verification; it does not establish a matched- 10^{-2} efficiency frontier or an optimal implementation of either classical method. In particular, a loose common upper witness cannot rank realized policy quality merely by the size of a certified upper bound. Section 16 reports the later method-specific witnesses. Their remaining error is not an omitted term that can be declared negligible.

TABLE XVI
HISTORICAL RESTRICTED STATE-SPACE CANDIDATES WITH A COMMON WITNESS

Method	Nodes/axis	Time steps	Regret bound	Generate (s)	Verify (s)
Markov chain	9	64	48.98350	0.12	3.84
Markov chain	17	128	48.95819	3.41	3.73
Markov chain	25	256	49.45469	42.90	3.74
Semi-Lagrangian	9	64	48.93352	0.08	3.67
Semi-Lagrangian	17	128	49.47658	1.06	3.69
Semi-Lagrangian	25	256	49.69852	25.90	3.74

All six policies are continuous bilinear state interpolants of bounded heads, piecewise constant in time, with the portfolio-distance factor. The common witness is the predeclared seed 16100 final critic, not a fitted value from either baseline. Generation discretizes the model; the separate certificate evaluates the continuous-time policy and unchanged stopping contract. None meets 10^{-2} , so this is not a matched-accuracy ranking.

21. ARITHMETIC INDEPENDENCE AND FRESH PRICE-CONTINUUM CONSTRUCTION

We distinguish two complete calculations. The first re-evaluates all eighteen historical policy/dual nodes with a new arithmetic implementation. Every basic operation and every exponential, logarithm, square root, and tanh endpoint is evaluated using 128-bit MPFR directed rounding, followed by directed conversion to binary64 endpoints. Stored network weights and policy constants are interpreted as exact dyadic inputs, while model decimals are enclosed as exact rationals. Invalid domains and division through zero fail closed. The kernel passes 450 exact-rational regression checks, but these checks are diagnostics rather than a formal proof of the implementation.

The full-node MPFR calculation recomputes stopped policy payoffs, dual source integrals, expenditure intervals, and the final exact rational envelope. It does not read historical certified endpoints as inputs. It does share the R14 analytic decomposition, rational Gauss-node isolation, derivative remainder lemmas, and stopping inequalities. Thus it is an independent established-arithmetic cross-check of the decisive calculation, not an independent discovery of every mathematical premise. Its complete price-uniform bound is 0.009969007708; full lower and upper intervals, not three pointwise comparisons, are retained at every node.

The second calculation starts from the economic model alone. It initializes the price set at $1/2$ and 8, generates a 16-slab feasible time-control policy at each node by deterministic SLSQP initialization, and fits the corresponding dual from deterministic starts. It then selects the worst exact current envelope breakpoint, rounded to a spacing of $1/1024$; a repeated node triggers bisection of the widest price interval. The declared stopping rule is a proved uniform gap below .01 or thirty freshly generated nodes. No inherited policy, dual pilot, or historically selected intermediate node is read. Optimizer attempts, unsuccessful intermediate envelopes, and resource use are retained.

The fresh pipeline reaches 0.009981645276 with 20 generated nodes. This result belongs to the time-control/dual method, not to the neural feedback. The fresh calculation uses the retained rational-Taylor interval kernel; the independent MPFR label belongs only to the separately re-evaluated eighteen-node object. Fresh generation closes a reproducibility distinction without removing the dual relaxation's accuracy floor. The original 16/32/64-slab experiment and its failure to attain .005 are retained below. Neither extra price nodes nor quadrature refinement alone supplies a convergence law to arbitrarily small error.

The historical R16 replication command executes that revision's neural trajectories and checks, both state-space solvers, the fresh price library, the historical-node MPFR audit, the structured inventory experiment, and the normalization calculations. Source identities, input

TABLE XVII
COMPLETE PRICE-CONTINUUM CALCULATIONS ON THE ORIGINAL ECONOMY

Calculation	Nodes	Uniform regret	Wall time (s)	Peak RSS (MiB)
Historical policies; MPFR	18	0.009969007708	1065.89	276.3
Fresh model-to-library	20	0.009981645276	415.79	497.5

Both cover $(t, u, x) = (0, 2, 1.25)$ and every $k \in [.5, 8]$. Historical inputs are frozen policies and dual pilots, not inherited endpoints; MPFR recomputes every node. The fresh calculation starts from model constants and deterministic initializations, with no historical policy or pilot. Its arithmetic is the retained rational–Taylor implementation; the MPFR label does not apply to the fresh library. Peak MPFR RSS is the largest worker, not summed concurrent memory.

hashes, complete cell records, failures, and hardware/software metadata accompany the results. Timings in different concurrent experiment processes are descriptive; they are not used to infer solver superiority from unequal accuracy or unequal processor contention.

21.0.0.1. *Historical evidence retained.* The following original-economy experiments, bounds, comparisons, and welfare-resolution calculations are the R14 research record. Their sources and results are unchanged. They are not additional R16 training runs or endpoints of the fresh R16 library.

22. RETAINED R14: EXECUTED NEURAL VERIFICATION ON THE ORIGINAL ECONOMY

The retained R14 neural experiment uses a state-feedback actor and critic with two hidden layers of 48 tanh units. Inputs are $(2t - 1, (u - 2)/.8, (x - 1.25)/.75)$. The actor’s output transformation is

$$c = .425 + .375 \tanh a_1, \quad \theta = .2 \tanh a_2, \quad p = .15 + .65 \tanh a_3.$$

The critic is $v(t, u, x) = G(u, x) + (1 - t)N(t, u, x)$, so its terminal trace is exact. The first-exit trace is not imposed by this architecture and must be checked.

All weights are freshly generated from seed 1401. Each actor-improvement step is preceded by three critic residual updates, with batch size 128. Adam learning rates are .0005 for the critic and .0002 for the actor, and the boundary-loss coefficient is .025. Snapshots at 100, 300, and 800 updates are retained. The experiment uses the 48-by-48 tanh architecture of the retained comparisons, but its continuous economic residual and boundary objective are a new adapter; it is not the same finite-step training objective as the external benchmark. The optimizer and sampled objective are proposal mechanisms only.

The certification program uses an independent interval implementation for network evaluation and differentiation. It checks all cells of $[0, 1] \times [1.2, 2.8] \times [.5, 2]$, all four first-exit faces, and the terminal face. A componentwise output neighborhood of radius 2^{-24} is included in the action ranges. This proves robustness for any admissible action within that neighborhood of the mathematical real-tanh network. It does not assert an undocumented error bound for a machine-library tanh routine or for a discretized diffusion simulator.

Table XVIII separates the stopping trace from the integrated policy residual and action gap. The finest cover has 16,384 interior cells and 512 cells on each stopping face. It gives the time-zero bound 23.493172463 and an all-initial-times bound below 23.656877772. The terminal trace is zero. The 800-update sampled residual mean square is approximately .0079446, but this does not control either the worst-case residual or the first-exit trace. Moreover, the 800-update

TABLE XVIII

WHOLE-DOMAIN NEURAL CERTIFICATES ON THE UNCHANGED ECONOMY, $k = 2$

Updates	Cells	$2b$	$\int e^{-\rho s}(2e + q)ds$	Regret upper	Seconds
100	2048	14.9778	8.8713	23.8491	4.40
300	2048	14.2970	9.0431	23.3400	4.75
800	256	14.2484	25.9010	40.1494	0.42
800	2048	14.1073	16.1408	30.2480	4.41
800	16384	14.0463	9.4470	23.4932	25.09

Upper bounds are rounded upward. Time is the measured verification stage, excluding training. Every row covers the complete domain with zero skipped and zero failed cells. The last three rows refine the cover for the same 800-update snapshot. None attains the .01 target.

TABLE XIX

STOPPING-TRACE ERRORS FOR THE FINEST NEURAL COVER

Stopping face	Cells	Trace error upper
$u = 1.2$	512	4.63971373
$u = 2.8$	512	7.02049755
$x = .5$	512	5.17780198
$x = 2$	512	7.02310795

Each entry is a uniform bound over the complete face and time interval. The comparison theorem pays the largest face bound twice. The error is an evaluation of the stored critic against $G - 8(1 - t)$, not a change in that settlement.

snapshot is not uniformly better than the 300-update snapshot at the same coarse cover. Both outcomes remain in the record.

The coefficient-transport calculation uses the same stored snapshot and cover, with

$$r \in [.019, .021], \quad \mu \in [.058, .062], \quad \sigma_u \in [.049, .051], \quad \sigma_x \in [.195, .205],$$

$$\rho \in [.039, .041], \quad k \in [1.9, 2.1],$$

and correlation fixed at $-.25$. Here μ denotes the risky-wealth drift coefficient multiplying p . The action bounds, state domain, and settlement are unchanged. Proposition 1 gives a time-zero regret bound below 23.530407814 uniformly over this model box. This demonstrates an executed transport mechanism beyond affine objective continuation. It does not make the base neural certificate accurate: the same boundary and approximation errors remain visible.

The finest computation records separate cellwise evaluation, action, and transport quantities. A diagnostic comparison of the interval neural derivatives with automatic differentiation checks 140 derivative components at 20 states and 60 actor components, with no discrepancy outside the numerical tolerance. This diagnostic detects implementation disagreement; it is not substituted for the full-cell interval proof.

23. RETAINED R14: INDEPENDENT NUMERICAL EVIDENCE AND ROBUSTNESS

The sharp economic comparison uses a second implementation that imports none of the inherited primal or dual certifiers. Its basic arithmetic encloses exact rational constants and exact stored binary64 inputs. Addition, multiplication, division, and reductions are rounded outward. Exponentials and logarithms are bounded by rational series with analytic remainders; tanh is

TABLE XX
INDEPENDENT CENTRAL-STATE CERTIFICATES FOR ALL RETAINED PRICES

k	Policy lower	Optimum upper	Regret upper	Policy width
0.5	-1.27266301	-1.26281817	0.009844825	1.13e-11
0.875	-1.27897331	-1.26925551	0.009717794	1.13e-11
1.25	-1.28479324	-1.27510918	0.009684054	1.13e-11
1.625	-1.29013398	-1.28045672	0.009677256	1.13e-11
2	-1.29501788	-1.28533017	0.009687701	1.13e-11
2.5	-1.30083961	-1.29112854	0.009711056	1.13e-11
3	-1.30590538	-1.29616492	0.009740453	1.13e-11
3.5	-1.31024866	-1.30047467	0.009773977	1.13e-11
4	-1.31389776	-1.30408950	0.009808244	1.13e-11
4.25	-1.31546968	-1.30564683	0.009822839	1.13e-11
4.5	-1.31688907	-1.30705332	0.009835742	1.13e-11
5	-1.31935183	-1.30949330	0.009858515	1.13e-11
5.5	-1.32141556	-1.31153758	0.009877966	1.13e-11
6	-1.32317048	-1.31327568	0.009894785	1.13e-11
6.5	-1.32468139	-1.31477191	0.009909467	1.14e-11
7	-1.32599609	-1.31607368	0.009922399	1.14e-11
7.5	-1.32715060	-1.31721673	0.009933858	1.14e-11
8	-1.32817262	-1.31822851	0.009944109	1.14e-11

Every upper endpoint covers all original adapted controls; every lower endpoint belongs to the identified feasible policy. The last column is the width of that policy's payoff enclosure, not its regret. Regret uses unrounded endpoints.

All 18 cases are included, and no inherited generation time is counted as zero.

obtained from the enclosed exponential. The Gauss nodes are isolated by exact polynomial signs and the weights are enclosed by their defining formula. The supplement states the floating-point model and the analytic remainder bounds, so these assertions are not based solely on rerunning a production function.

The independent primal evaluator uses 262,144 integrand evaluations per 16-interval policy. The fine dual uses 262,144 source boxes per price. Complete supporting-plane covers certify a negative second-derivative upper bound below $-.06717$ on the first preference region and a strict derivative separation on the remaining region. They also certify a source floor above -6.98956 , a source ceiling below zero, and an absolute preference coefficient below $.97119$. These independently computed margins imply the analytic bounds $S_0 \geq -7.1$ and $|\beta| \leq 1.1$ used for stopping continuation. The complete old policy and dual inputs are retained and identified by their content hashes.

Exact price postprocessing is independently implemented by enumerating affine intersections with rational arithmetic, rather than importing the original hull routine. Its input expenditures are newly computed from the stored preference actions with an explicit stopping correction. Thus the upstream policy values, upper values, expenditure intervals, and downstream envelope all have a new execution path. Reproducibility of this path remains distinct from independence of its analytic premises.

23.1. Stress tests for the entire library

The robustness experiment changes every dual pilot: y_0 is increased by 10^{-6} , and its reference preference drifts receive alternating perturbations of $\pm 10^{-6}$ with inward projection into the admissible reference interval. Simultaneously, the proof uses the weaker source floor -7.11 , the

TABLE XXI
SEPARATE QUANTITIES IN THE INDEPENDENT $k = 4.25$ CALCULATION

Quantity	Lower	Upper
Source integral	-2.38605836997	-2.38604817428
Covariance	-0.00017879144	-0.00016944433
Reference-control gap	0	0.00003256016
Conditional variance	0.00003697298	0.00003697299
Localization	2.4594E-7	2.4595E-7
Primal payoff	-1.315469674846	-1.315469674834

Intervals are displayed outward. The source-remainder enclosure, Gaussian tail, terminal tangent, reference cost, and arithmetic allowances are present in the machine-readable record. The conditional-variance allowance is positive and is not removed when quadrature is refined.

TABLE XXII
CONTINUUM CERTIFICATES UNDER INDEPENDENT VERIFICATION AND PERTURBATION

Certificate	Uniform regret upper
R12 retained implementation, original inputs	0.009994024847
R14 independent implementation, finer cover	0.009969007711
R14 perturbed witnesses and enlarged allowances	0.009970977450
R14 joint state–cost rectangle	0.012095431554

The first three rows refer to the central initial state. The fourth covers the additional initial-state rectangle K and is not compared with the central-state .01 tolerance as though the domains were identical. The perturbation experiment changes proof witnesses and allowances, not the economic primitives.

larger coefficient bound 1.11, twice the Gaussian-tail allowance, and 1.1 times the conditional-variance allowance. The resulting continuation floor remains strictly above -8 . All 18 upper certificates are recomputed, while the original economy and feasible lower policies remain unchanged.

The unperturbed independent bound leaves approximately 3.10×10^{-5} below .01, compared with approximately 5.98×10^{-6} for the retained R12 bound. The perturbed calculation remains below .01. Neither statement claims a bound below .009. An additional exact sensitivity calculation shows how a common enlargement of every upper and lower endpoint changes the final envelope: enlarging each side by δ increases the guaranteed gap by at most 2δ . It is a sensitivity result conditional on valid node intervals, not a replacement for the independent primitive audit.

A third, materially different decomposition integrates the capped and interior conjugate using closed truncated-lognormal moments. At prices .5, 4.25, and 8, a separate program computes the source integral, initial affine coefficient, and covariance term at 40 and 70 decimal digits. Every computed quantity lies inside the independently certified interval. The agreement provides a useful cross-check of algebra and decomposition; these point-valued high-precision quadratures are not themselves rigorous enclosures, and their convergence is not used as a proof of the interval certificate.

24. RETAINED R14: ACCURACY, COMPUTATIONAL WORK, AND COMPARATORS

A certification time measured from frozen proof objects is not the cost of generating those objects. The 18-node independent audit takes approximately 103.08 seconds of measured wall time in the retained run, but starts from existing policies and dual pilots. The 72-corner state

TABLE XXIII

FRESH GENERATION AND INDEPENDENT VERIFICATION AS POLICY RICHNESS INCREASES

Slabs	Generation	Dual fit	Primal audit	Dual audit	Regret upper
16	0.235	0.284	0.737	7.408	0.009688058
32	0.364	0.434	0.926	6.815	0.009669548
64	0.847	0.379	1.791	5.748	0.009665074

All stage times are seconds. Each case starts without inherited policy or dual inputs, and all three optimizations report success. End-to-end measured case times are 8.67, 8.54, and 8.77 seconds. The experiment changes policy richness, unlike a price-node refinement experiment. All cases pass .01; none passes .005, .0025, or .001.

audit takes approximately 29.71 seconds. These are verification costs. The generation costs of inherited and interrupted historical attempts are not reconstructed from their absence and are not assigned zero.

The fresh neural run records all 800 actor updates and 2,400 critic updates, the fixed seed, every retained checkpoint, and resource usage. Its recorded execution after program entry is approximately 13.08 seconds, with interpreter and import time separately recorded. It uses one Torch CPU thread and no GPU. The execution environment reports an AMD EPYC 9V74 processor and five exposed logical CPUs; this identifies the environment, not exclusive ownership of a physical machine. Stage times recorded during overlapping local experiments are descriptive clocks, not controlled hardware-efficiency comparisons.

24.1. *A newly generated policy-richness frontier*

The same original economy at $k = 2$ is also solved within deterministic time-control classes of 16, 32, and 64 intervals. Each problem starts from a fixed feasible constant-consumption construction and preference action .05. Constrained sequential quadratic optimization uses analytic gradients to propose a policy. A separately initialized scalar-deflator and reference-drift fit proposes its upper witness. The fresh policy is then checked by the independent stopped evaluator and the fresh upper witness by the independent dual. The fitting objective is not counted as a certificate.

The frontier is informative because its approximation object is held distinct from its upper relaxation. Policy refinement improves the lower bound, but the independently fitted affine upper bound is nearly unchanged. A fixed specialized relaxation can therefore limit the attainable regret even when quadrature error is very small. This calculation is not evidence of convergence to arbitrary tolerance, and it does not establish convergence of the neural optimizer.

The classical comparator is a freshly executed direct time-control optimization. It is a serious feasible-policy and upper-bound calculation on the same original economy, but it is not two distinct state-space HJB methods and it does not furnish a matched-accuracy advantage for the neural method. A comparative claim requiring monotone PDE, semi-Lagrangian, or controlled Markov-chain baselines at the same certified accuracy remains outside the completed experiment. Likewise, the preserved Riccati-reference study used structured gain outputs; its errors cannot be relabeled as errors of the present 48-by-48 tanh actor and residual-training pipeline.

24.2. *Retained comparisons and high-dimensional scope*

The historical SOC-MartNet holdouts, method-specific tuning studies, and clipped-feedback comparator are preserved without suppressing unfavorable panels. The clipped-feedback comparator's lower mean cost and the absence of a robust ordering in several higher-dimensional

TABLE XXIV
CERTIFIED WELFARE LOSSES WHEN THE ADJUSTMENT PRICE RISES

p	q	Lower loss	Upper loss	Strict	Relative
0.5	2	0.01266717	0.02940793	Yes	No
2	8	0.02321063	0.04284245	Yes	Yes
4	4.1	0E-8	0.00196053	No	No
4	8	0.00433075	0.02408312	Yes	No
6	8	0E-8	0.01489694	No	No

“Strict” means a strictly positive lower endpoint. “Relative” means interval width no greater than that endpoint. A “No” in the strict column does not reverse the economic monotonicity theorem; it means that strict loss is not separated from zero by these intervals.

seed panels limit what can be inferred about candidate-generation quality. They do not invalidate the stopped comparison theorem, just as the theorem does not establish empirical superiority.

The 128-dimensional experiment in the retained record certifies a rank-one-coupled Hamiltonian action oracle at frozen jets. Its nonconvex coupling can be reduced to a scalar aggregate search with strongly convex conditional subproblems. This is an action-optimization result. It is not a certified solution of a 128-dimensional dynamic economic model. The actual neural state-domain calculation in this revision concerns the two-state economy in Section 2; no action-dimension result is used to inflate that state dimension.

25. RETAINED R14: WHAT THE CERTIFIED COUNTERFACTUALS IDENTIFY

For any policy, raising k subtracts $(q - p)B(\pi)$, where

$$0 \leq B(\pi) = \mathbb{E} \int_0^\tau e^{-\rho s} \theta_s^2 / 2 \, ds \leq .02 \frac{1 - e^{-\rho}}{\rho}.$$

Consequently V_k is nonincreasing and convex. Monotonicity determines a weak sign without numerical computation, but not a strictly positive or economically well-resolved welfare loss.

Let $[L(k), U(k)]$ be the independently certified central-state value envelope. For $p < q$, a valid loss interval is

$$[\max\{0, L(p) - U(q)\}, \min\{U(p) - L(q), .02C_\rho(1)(q - p)\}], \quad C_\rho(h) = \frac{1 - e^{-\rho h}}{\rho}. \quad (62)$$

A positive lower endpoint establishes strict loss. We report a separate relative-resolution criterion: the width of this interval is no greater than its positive lower endpoint. Neither criterion follows automatically from a .01 bound on policy regret.

The contrast between $2 \rightarrow 8$ and $4 \rightarrow 4.1$ is substantive. The first comparison is strictly positive and relatively resolved by the current intervals. The second has a small absolute upper bound but no certified positive lower bound from interval separation. A policy may therefore be sufficiently accurate for an absolute deployment tolerance while the current value bounds remain too wide for a local comparative-static statement.

The policy library also has a direct economic interpretation. Its selector chooses a stored control vector on each exact price region; within that region no new optimization is required. The supplement lists the selected regions, their widths, consumption ranges, and average preference adjustments. These are properties of feasible chosen policies. They are not claims that the optimal policy itself is piecewise constant in the price, or that policy expenditures equal derivatives of the unknown optimum.

TABLE XXV
PRICE REGIONS IN WHICH WELFARE LOSSES ARE SEPARATED

Anchor p	Strictly positive loss	Relative width at most one
0.5	(1.093292, 8.0]	(2.617755, 8.0]
2	(2.886123, 8.0]	(6.450558, 8.0]
4	(6.207877, 8.0]	—
6	—	—

Decimal endpoints are descriptive approximations of exact rational thresholds, not proof endpoints. The exact fractions and open-endpoint convention are in the price-audit file. A dash means no such region up to price eight is established by this predicate. The relative criterion here uses the untruncated difference of value intervals and is therefore conservative.

The tolerance .01 is an absolute loss criterion in the utility units of (5). It is not a scale-free welfare measure. The relative and strict-loss diagnostics in (62), rather than the target alone, determine which of the reported economic comparisons are resolved. No empirical calibration or external welfare normalization is imposed to make this declared numerical target appear economically universal.

25.1. A consumption-top-up interpretation

An absolute utility tolerance is not an economic unit. We therefore define a compensation experiment before reporting it. Keep the policy's preference, wealth, adjustment cost, and stopping paths fixed, and provide an externally financed proportional consumption top-up $\lambda \geq 0$. Only the consumption entering instantaneous utility changes, from c_s to $(1 + \lambda)c_s$. This is an in-kind welfare comparison; it is not a claim that the extra consumption is financeable under the unchanged budget or the original consumption cap.

PROPOSITION 13—A scale-normalized compensation bound: *For a feasible time-control policy with $c \leq 1$, $u \in [1.2, 2.8]$ until stopping, suppose $V - J^\pi \leq \epsilon$ and $\mathbb{E} \int_0^\tau e^{-.04s} ds \geq A > 0$. If $1 - 1.8\epsilon/A > 0$, a top-up no larger than*

$$\bar{\lambda} = (1 - 1.8\epsilon/A)^{-1/1.8} - 1 \quad (63)$$

is sufficient to compensate the certified policy loss in the specified welfare experiment.

For the fresh central-state price library, the rigorous original stopping calculation supplies

$$A \geq \frac{1 - e^{-.04}}{.04} \left\{ 1 - 2 \exp\left(-\frac{.58^2}{2(.05)^2}\right) \right\}.$$

The resulting uniform top-up bound is 1.033008 percent. It is not assigned to a neural policy. Under a common positive rescaling of the entire welfare criterion, both the loss and the marginal utility gain rescale together, leaving this compensation experiment unchanged. This does not supply an empirical calibration or a wealth-equivalent claim. The exact definition prevents a numerical threshold from being presented as economically meaningful merely because it is slightly below .01.

26. BUDGET-FEASIBLE WELFARE COMPENSATION IN THE ORIGINAL ECONOMY

An absolute utility tolerance is not self-interpreting. The consumption-top-up experiment in the preceding section specifies an external financing convention, but it is not a budget-feasible wealth equivalent. We now give a sufficient initial-wealth compensation with an explicit feasible policy adjustment under the original budget and stopping rules. The object being compensated is the sharp deterministic time-control library, not the original neural policies.

Let $\pi = (c, \theta, 0)$ be the selected library policy at $(u_0, x_0) = (2, 1.25)$ and an arbitrary $k \in [.5, 8]$. Add initial wealth Δx and a constant consumption increment

$$d_c = \frac{\Delta x}{C_{.02}(1)}, \quad C_{.02}(1) = \frac{1 - e^{-.02}}{.02},$$

while retaining the preference control and zero portfolio. Before preference stopping, the wealth difference solves $\dot{z} = .02z - d_c$, $z(0) = \Delta x$, and has $z(1) = 0$ and $z(s) \geq 0$. No extra payoff, transfer after time zero, or modified consumption cap is introduced.

PROPOSITION 14—Sufficient wealth compensation: *Suppose every selected original and compensated policy avoids wealth stopping before time one, satisfies $c + d_c \leq .8$, and has preference-exit probability at most p_e . Then*

$$J_k(0, 2, 1.25 + \Delta x; \pi^+) - J_k(0, 2, 1.25; \pi) \geq d_c \cdot 8^{-1.2} C_{.04}(1)(1 - p_e). \quad (64)$$

Consequently, if the right side exceeds the original library's uniform regret bound ε_{lib} , the compensated feasible policy satisfies $J_k(0, 2, 1.25 + \Delta x; \pi^+) \geq V_k(0, 2, 1.25)$ for every library-covered price.

The comparison is sufficient compensation with an explicit policy change. It is neither the exact minimum wealth equivalent of the unchanged feedback nor a calibrated empirical welfare estimate.

Directed calculations check every one of the 20 freshly generated library policies. The preference-exit bound is below 1.207×10^{-29} . The analytic sufficient wealth increment is at most .007713143; the implementation chooses the simple rational increment $39/5000 = .0078$, or .624% of initial wealth. Its consumption increment is enclosed by $[\text{.007878259998266659}, \text{.007878259998266711}]$. All compensated consumption paths satisfy the original .8 cap, all terminal wealth values remain strictly above .5, and the global wealth upper bound is below $1.257958 < 2$.

The resulting uniform utility gain is at least .010094048517141695, greater than the original uniform regret upper bound .009981645275150275, with a verified slack exceeding .00011240324. An additional replay through the original stopped-payoff evaluator is retained separately. Thus the normalization is tied to a feasible economic experiment rather than to the numerical size of .01 alone. Its coverage remains the original library's central initial preference and complete price interval, not the full-state neural experiment.

27. CONCLUSION

The new stochastic policies establish three different facts with different tests: strict expert-payoff improvement, uniform initial-to-final mixture improvement, and useful optimality bounds on a continuum of initial states and adjustment prices. The Jensen-gap argument is needed for the second fact; it cannot be replaced by four successful expert updates. The martingale-price

decoder is needed for admissibility; it cannot be replaced by a sampled wealth trajectory. The unrestricted dual is needed for optimality; it cannot be replaced by policy comparisons alone.

These additions advance the original neural computation program without discarding its difficult evidence. They concern newly trained, genuinely stochastic neural policy generators, not a renamed deterministic library. They also retain the strong direct stochastic comparator, which supplies similarly sharp local accuracy without hidden layers and without a demonstrated neural speed disadvantage being concealed. The older complete-domain feedback certificates, continuation-trace obstruction, and high-dimensional correction results keep their stated scopes.

The all-domain .01 feedback objective is unchanged. Extending the present domain requires an admissible policy system and a sharp comparison object covering additional initial states and starting times, together with a matched global classical frontier. The present manuscript supplies complete, reproducible regional theorems and calculations for that program; it does not replace the global objective with an assertion that it has already been attained.

APPENDIX A: PROOFS FOR THE STOCHASTIC POLICY CONSTRUCTION

A.1. Self-financing and admissibility

For each slab, the martingale-measure backward equation for (14) is

$$F_t + .11yF_y + .045y^2F_{yy} - .02F + c = 0, \quad F(1, y) = R.$$

Applying Itô's formula under the physical measure gives

$$dF = (.02F - c - .09yF_y)dt - .3yF_y dW^x.$$

Substitution of $p = -1.5yF_y/F$ yields exactly the original wealth equation. Continuity of the conditional price at the finitely many slab boundaries eliminates any jump or unfinanced reset. Formula (13) makes $F(0, 1) = x$ exactly.

Let $a(y) = c(y) - .01$. Since $a \in [.49, .79]$,

$$F(t, y) = .5 + (R - .5)e^{-.02(1-t)} + \int_t^1 e^{-.02(s-t)} \mathbb{E}^Q[a(s, Y_s) | Y_t = y] ds.$$

This implies $.5 < F < X_{\max}$, where

$$X_{\max} = .5 + .79 \frac{1 - e^{-.02}}{.02} + .0001 < 1.282253.$$

For $q = \sigma(b - s \log y)$, elementary maximization over $q \in [0, 1]$ gives

$$\frac{|yc_y|}{c - .01} = s \frac{.3q(1-q)}{.49 + .3q} \leq s\kappa, \quad \kappa = \frac{\sqrt{.79} - \sqrt{.49}}{\sqrt{.79} + \sqrt{.49}}.$$

The conditional factor is multiplicative in y , so differentiation under its bounded integrand gives

$$|yF_y| \leq 6.5\kappa(F - .5), \quad F_y \leq 0.$$

Consequently $0 \leq p \leq 1.5(6.5)\kappa(1 - .5/X_{\max}) < .706888$. The deterministic preference drift and consumption satisfy the original action constraints. These arguments establish Theorem 2 for every $y > 0$, not merely inside a quadrature truncation.

A.2. Concavity, common stopping event, and continuum accuracy

Write $r_u = u - 1 \in [.2, 1.8]$ and $a = -\log c \in [-\log .8, \log 2]$. The diagonal entries of the Hessian of $f = -e^{r_u a} / r_u$ are negative. Its determinant has the sign of

$$D(r_u, a) = r_u^2 a^2 - 2r_u(r_u + 1)a + 2(r_u + 1).$$

Since $a < .7$, D decreases in a . The polynomial $D(r_u, .7)$ is concave in r_u , so its minimum is at an endpoint. At $r_u = 1.8$ it is $.1316 > 0$ and at $r_u = .2$ it is larger. Hence f is jointly concave on the full stated rectangle. Both $-k\theta^2/2$ and $G(u, x) = -.02(u - 2)^2 + .1 \log x$ are concave for $k \geq 0$.

All expert preference drifts are in $[0, .2]$. On

$$E = \{\sup_{t \leq 1} |.05W_t^u| < .58\},$$

all four expert preference processes, and their convex mixture, stay inside $(1.2, 2.8)$. There is no wealth exit by Theorem 2. The exponential maximal inequality gives $\Pr(E^c) \leq p_E$. On E , running and settlement concavity give the required pathwise mixture lower bound up to the common terminal time. For these particular admissible policies, the original stopped payoff lies in $[-15, 1]$ for all $k \in [.5, 8]$: running utility has magnitude below six, the adjustment penalty is at most $.16$, and the remaining stopping settlement fits in this conservative range. The error on E^c is at most $16p_E$. This proves (18). Convexity of the explicit dual bound gives

$$\bar{V}_k(z) \leq \sum_i \lambda_i \bar{V}_k(z_i),$$

and proves (19). The argument uses neither concavity nor a numerical approximation of V^* .

A.3. An upper bound for the initial mixture's Jensen gain

The stored initial experts have identical slope and preference-drift vectors within each seed; this is checked exactly from their serialized coefficients. The four consumption intercepts differ only by budget offsets. Thus, in slab j ,

$$d_{c,j} := .3 \tanh\{(\max_i b_{ij} - \min_i b_{ij})/4\}$$

bounds the full-factor range across the four consumptions. The preference range is $d_u = .04$ at every time. No loss in this bound is hidden by testing a few factor values.

On the event $|.05W_t^u| \leq .2$, all four preferences belong to $[1.78, 2.42]$. Directed interval evaluation on a 64-by-64 cover of this rectangle times $[.5, .8]$ gives

$$|f_{uu}| \leq 4.436941, \quad |f_{uc}| \leq 3.709523, \quad |f_{cc}| \leq 25.902278.$$

These are enclosing Hessian evaluations over boxes, not derivative samples. For any four nonnegative weights summing to one, weighted variance of a quantity in a range of width d is at most $d^2/4$. Taylor's theorem about the weighted mean and Cauchy-Schwarz therefore bound the flow Jensen gap by

$$\gamma_j = \frac{H_{uu}d_u^2 + 2H_{uc}d_u d_{c,j} + H_{cc}d_{c,j}^2}{8}.$$

The complement of the marginal event has probability at most $2\phi(4)/4 = e^{-8}/(2\sqrt{2\pi})$, uniformly for $t \leq 1$. Its flow Jensen gap is at most six on E . The initial effort Jensen gap is zero because all preference drifts coincide. If d_R bounds the range of exact terminal reserves, the terminal Jensen gap is at most

$$e^{-.04} \left(.005d_u^2 + \frac{.0125}{.5^2} d_R^2 \right).$$

Integrate γ_j using the exact discounted slab masses, add six times the marginal-tail bound integrated over the horizon, add the terminal term, and add $16p_E$ for E^c . The resulting directed upper endpoint is Γ_0 . Each reserve range is formed from its entire enclosing interval. This proves the premise of Theorem 4. Subtract the old upper bound from the final lower bound to obtain (20). If final regret is $e(z) \leq B$, old regret is at least $e(z) + d$. Since $e/(e + d)$ increases in e , (21) follows. In particular, a zero final regret creates no division problem because $d > 0$ bounds the old regret away from zero.

A.4. Conditional Gaussian evaluation and analytic error bounds

The independent payoff checker does not import the neural optimizer or simulate paths. Put $t = v^2$ and represent $\log Y = -.025t + .3vz$, with z standard normal. Conditional on z , preference minus one has mean

$$r_0 = u_0 - 1 + \int_0^t \theta_s ds + .0125vz$$

and independent Gaussian variance $.00234375t$. The sign of the conditional mean reflects the original negative correlation and the convention $z = -W_t^x/\sqrt{t}$.

On $|z|, |z_2| \leq 8$, the checker has $r_0 \geq .88$, actual $u - 1 \geq .49$, actual $u - 1 \leq 1.71$, and $|\log c| < .7$. Outside the original preference interval, define a bounded utility extension by clipping preference to $[1.2, 2.8]$. The extension is an integration device only: it is never a change to the stopped economic model, whose difference is bounded below. Expand conditional utility to degree ten in the independent Gaussian increment. Its first omitted even moment gives the uniform remainder

$$10(10395) \frac{(.00234375)^6}{(.49)^{13}} < 1.836 \times 10^{-7}.$$

For completeness, the derivative bound follows from

$$\frac{e^{ar}}{r} = \int_{-a}^{\infty} e^{-ry} dy, \quad r > 0,$$

by bounding its twelfth derivative divided by $12!$. The integral on $[0, \infty)$ equals r^{-13} , and the integral on $[-a, 0]$ is bounded by $e^{ra}a^{13}/13!$. The conservative multiplier ten bounds their sum on the stated ranges.

Let $T_0 = 2\phi(8)/8$ and $T_{2j} = 28^{2j-1}\phi(8) + (2j-1)T_{2j-2}$. These bound the omitted even Gaussian tail moments. The conditional-polynomial correction is

$$\sum_{j=0}^5 \frac{10(.00234375)^j}{.88^{2j+1}} T_{2j},$$

and $12T_0$ bounds the two omitted bounded-utility tails. The price integral and its derivative have separate normal-tail bounds $.8T_0$ and $1.95T_0$. This calculation never integrates the singular unextended CRRA expression over an unrestricted Gaussian preference distribution.

Each transformed time slab is subdivided to width at most $1/40$ in v . The normal interval $[-8, 8]$ has 96 cells. Both integrals use degree-15 exact, eight-point Gauss rules, with rationally isolated nodes and outward weights. The inherited analytic rule bounds its remainder on a complex disc of radius $1/16$ in v and $1/2$ in z . Here is an explicit justification for its common modulus bound $M = 100$. On these discs the imaginary logistic arguments are respectively bounded by

$$6.5\{(2(.065) + 2.4)/16 + .065/256\} < 1.03, \quad 6.5(.3)/2 = .975 < \pi/3.$$

Hence the sigmoid has no pole, its real part lies between zero and one, $\Re c \geq .5$, $|c| \leq .8$, and $|\log c| < .8$. Also $\Re r_0 > .8$, $|r_0| < 1.4$, and the complex independent variance has modulus below $.0027$. The conditional Taylor polynomial is bounded by

$$\sum_{j=0}^5 \frac{10(2j-1)!!(.0027)^j}{.8^{2j+1}} < 12.554.$$

The coefficient estimate uses $e^{2(1.4)(.8)} < 10$. Multiplication by $|2v| \leq 17/8$, the discounted density bounds, and the Gaussian density modulus below $.5$ leaves a bound below fifteen. The price and price-derivative integrands are smaller. Thus $M = 100$ is conservative for each analytic integrand. The quadrature remainder is charged separately from arithmetic enclosure width and from Gaussian truncation. The two arithmetic implementations may choose different outward time subdivisions; they need not have identical node counts.

Finally, the full-horizon extended-flow calculation differs from the original stopped payoff only after preference exit. With

$$M_4 = .22^4 + 6(.22)^2(.0025) + 3(.0025)^2,$$

the correction $20p_E + .02\sqrt{M_4 p_E}$ bounds the bounded running and stopping terms and the unbounded quadratic terminal-preference term by Cauchy–Schwarz. Terminal log wealth uses the enclosed exact reserve. Each reported payoff interval includes all these errors. Independent MPFR arithmetic checks the same analytic derivation, rather than proving an unrelated theorem.

A.5. Price transport and welfare transfer

For fixed policies τ is independent of k . The adjustment integral is affine in k and the omitted cost after stopping is at most $.02C_{\rho}p_E$, proving (22). The dual-node chords are upper bounds because the unrestricted value is a supremum of affine functions of k . Apply the appropriate sign of $k - 2$ to the payoff-cost interval. On every dual-node segment the resulting vertex upper-minus-lower bound is affine; its endpoint maximum bounds the whole segment. State mixing then proves (23).

For compensation, apply Theorem 3 to expert initial wealths $x_i + .002$, with the same weights for the original z . The actual initial wealth is $x + .002$. Rechecked reserves and slopes retain Theorem 2. Compare the resulting policy lower bound with the convex dual upper at the original, uncompensated z . The smallest directed vertex margin, less $16p_E$, is $0.00004475 > 0$. This proves the sufficient initial-wealth compensation without interpreting a utility gap as an unfinanced consumption gift.

APPENDIX B: PROOFS OF THE VERIFICATION AND TRANSPORT RESULTS

PROOF OF THEOREM 1: For any admissible control a , stopped Itô's formula gives

$$J(t, z; a) - v(t, z) = \mathbb{E}_{t, z} \left[\int_t^\tau e^{-\rho(s-t)} \mathcal{R}_{a_s} v(s, Z_s) ds + e^{-\rho(\tau-t)} (g - v)(\tau, Z_\tau) \right].$$

One may first stop the local martingale at a bounded localization sequence. Bounded derivatives and coefficients on the stopped compact domain give integrability, and dominated convergence then yields the displayed identity. For the candidate π , the absolute value of the right side is at most $b + \int_t^\tau e^{-\rho(s-t)} e(s) ds$. For an arbitrary original control, the residual is at most $e(s) + q(s)$, because its action is covered by the continuous maximum in (8). Hence

$$V(t, z) - v(t, z) \leq b + \int_t^\tau e^{-\rho(s-t)} \{e(s) + q(s)\} ds.$$

Adding the lower-policy error proves (9). Nonnegativity follows from admissibility of π . If an implemented action belongs to an enclosed neighborhood, the same residual and action-gap checks hold for it by interval inclusion, so the proof is unchanged. No statement about the convergence of the process that generated v or π is used. *Q.E.D.*

PROOF OF PROPOSITION 1: In the perturbed model, the candidate residual is at least $-e - d$, while the residual of any original admissible action is at most $e + q + d$. Applying the stopped identity separately in that model bounds the lower-policy and upper-value errors by the corresponding integrals. Their sum is $2e + q + 2d$, not an extra improvement-gap penalty on top of those errors. Nonnegative integrands permit replacement of the perturbed discount by ρ . The stopping-trace error is at most $b + b_{\text{par}}$ if the settlement differs by at most b_{par} . This argument neither equates the two stopping times nor assumes a pathwise coupling of different diffusions. *Q.E.D.*

APPENDIX C: PROOF OF THE JOINT STATE-PRICE CONSTRUCTION

LEMMA 1—Concavity of the covered policy functional: *For each stored policy satisfying (48), there is a concave function $F_j(z; k)$ of $z \in K$, affine in k , such that $J_k(0, z; \pi_{j,x}) \geq F_j(z; k)$. Its values at the four corners have the independently computed lower endpoints used in the state-price calculation.*

PROOF: Put $r = u - 1$ and $L = -\log c$. The determinant expression in Section 15 has the sign of

$$r^2 L^2 - 2r(r+1)L + 2(r+1) = r^2 L^2 + 2(r+1)(1 - rL) > 0,$$

while $f_{cc} < 0$. Therefore $D^2 f$ is negative definite over the stated region. In the covered Gaussian integral, the preference input is affine in initial preference and consumption is affine in initial wealth. Their image remains in the region just verified. Integrating this concave composition with positive Gaussian and time weights preserves concavity.

The full-horizon preference settlement is the concave quadratic $-.02\{(u - 2 + \int \theta)^2 + .0025\}$. By the budget shift the terminal wealth, and hence its logarithmic settlement, is independent of initial wealth. The deterministic expenditure is independent of the initial state. Subtract a uniform normal-tail and stopping allowance from this covered integral plus its terminal settlement and expenditure. The supplement proves this allowance for all $z \in K$ and $k \in [.5, 8]$. The resulting

exact function is F_j , which retains concavity and affine dependence on price. Its coefficient of $-k$ is

$$B_j^{\text{full}} = \frac{1}{2} \int_0^1 e^{-\rho s} \theta_{j,s}^2 ds.$$

At each corner the interval quadrature encloses the exact covered integral and terminal term. Its lower endpoint after the uniform corrections is no larger than F_j there. The finite quadrature rule is not assumed to reproduce the exact functional, and sampled corner values are not used without their enclosure errors. *Q.E.D.*

PROOF OF PROPOSITION 10: Let the four corners of K be z_v , and write $z = \sum_v \lambda_v(z) z_v$ using the nonnegative bilinear corner weights, which sum to one. Convexity of the verified upper object and concavity of the lower functional imply

$$V_{k_i}(0, z) \leq U_i(z) \leq \sum_v \lambda_v(z) U_i(z_v), \quad J_k(0, z; \pi_{j,x}) \geq F_j(z; k) \geq \sum_v \lambda_v(z) F_j(z_v; k).$$

For k between k_i and k_{i+1} , fixed-policy affinity in price and a common admissible policy set give convexity of V_k . Applying the value chord at the two endpoints yields the chord between the two verified upper corner combinations.

Suppose a policy lower corner endpoint is L_{jv} at k_j and its expenditure belongs to $[B_j^-, B_j^+]$. Its sign-aware lower line is

$$L_{jv} - (k - k_j) B_j^+ \quad (k \geq k_j), \quad L_{jv} - (k - k_j) B_j^- \quad (k \leq k_j).$$

Each price cell between consecutive nodes has a fixed sign for every policy. The upper chord minus this lower line is therefore an affine function $g_{jv}(k)$. For every state, the regret of policy j is at most $\sum_v \lambda_v(z) g_{jv}(k) \leq \max_v g_{jv}(k)$. Selecting a policy minimizing this last bound proves the uniform state guarantee

$$R(k) = \min_j \max_v g_{jv}(k).$$

Take all pairwise intersections of the finitely many affine functions and the price-cell endpoints. Between successive such points, their ordering does not change. The maximum for each policy and then the minimum across policies select a single affine function on that interval. Its maximum occurs at an endpoint. Exact rational evaluation at this finite set thus bounds the supremum over all real prices. A corner line is discarded only when another corner line from the same policy dominates it at both price-cell endpoints, which proves domination on the entire cell. The program records those reductions and checks the remaining intersections without a sampled-grid extrapolation. *Q.E.D.*

PROOF OF PROPOSITION 11: The independent primal and dual constructions in the supplement supply, respectively, valid original-policy lower endpoints and upper functions (47). Exact action and terminal-budget checks hold for every central policy and every shifted policy at the extremes of the initial-state rectangle. Affinity of the shift extends these feasibility checks to its interior. Lemma 1 supplies the concave lower functional with a common tail and stopping allowance. All quantities in Proposition 10 are therefore present.

The execution uses the 18 prices

$$.5, .875, 1.25, 1.625, 2, 2.5, 3, 3.5, 4, 4.25, 4.5, 5, 5.5, 6, 6.5, 7, 7.5, 8.$$

The stored state–price record checks 211 candidate breakpoints and returns the exact fraction displayed following (50); upward decimal conversion gives (49). The central-state calculation uses the exact stopped-policy expenditure intervals and checks 187 candidate breakpoints with a separate all-intersections implementation. Its exact worst-case fraction is

$$\frac{1600901779824582944359922627717}{160587876576305697877131733762048},$$

whose upward display is 0.009969007711. These are computer-assisted conclusions conditional on the explicitly stated arithmetic model and analytic enclosure lemmas, not consequences of agreement between two floating-point approximations. *Q.E.D.*

APPENDIX D: ECONOMIC RESOLUTION AND THE COMMON-ENDPOINT STRESS BOUND

For $p < q$, every policy loses $(q - p)B(\pi) \geq 0$, and $B(\pi) \leq .02C_\rho(1)$. Taking suprema preserves monotonicity and gives $0 \leq V_p - V_q \leq .02C_\rho(1)(q - p)$. Combining this fact with the independently verified intervals at p and q gives (62). On any segment where the upper and lower envelope pieces are affine, both strict separation and the conservative relative-width inequality reduce to affine inequalities in q . Their roots, included with the exact envelope breakpoints, produce the resolution regions without a grid test.

Finally, if every nodal upper endpoint is enlarged by δ and every nodal lower endpoint is decreased by δ , every upper chord is enlarged by δ and every lower policy line is decreased by δ . Their maximum/minimum envelope gap increases by at most 2δ . This proves the recorded endpoint stress test and also its scope: it tests sensitivity to valid endpoint inflation, not the validity of upstream node proofs.

APPENDIX E: PROOFS FOR STOPPING ACCESSIBILITY AND CERTIFIED ITERATION

PROOF OF PROPOSITION 6: Fix $x_0 \in [1.5, 2)$ and use $c = .8$, $\theta = p = 0$. Wealth decreases, stays below two, and satisfies $x_s \geq x_0 - .8s \geq .7$ for $s \leq 1$. Thus it does not hit either wealth face. Preferences equal $2 + .05W_s$. On the event $H = \{\sup_{s \leq 1} |.05W_s| \leq .2\}$ there is no preference exit, $u \in [1.8, 2.2]$, and running utility is at least -1.5 . To see the last inequality, set $r = u - 1 \in [.8, 1.2]$. The function $-(5/4)^r/r$ is increasing on this interval since $r \log(5/4) < 1$. Its value at $.8$ exceeds -1.5 , using $(5/4)^4 < (6/5)^5$. Settlement on H is at least $-.05$, because $-.02(.2)^2 + .1 \log(.7) > -.05$. Discounting negative terms cannot worsen these lower bounds. The pathwise discounted payoff on H is therefore at least -1.55 .

Until the original stopping time, $u \in [1.2, 2.8]$. The same elementary monotonicity bounds running utility by -6 from below, and $G \geq -.1$. Even on H^c , the stopped payoff is at least $-6 - 8 - .1 = -14.1$. The Brownian maximal inequality gives $\Pr(H^c) \leq 2e^{-8}$. Hence

$$J_2(0, 2, x_0; .8, 0, 0) \geq -1.55 - 12.55(2e^{-8}).$$

Subtracting $g(0, 2, 2) = .1 \log 2 - 8$ proves (30), uniformly as $x_0 \uparrow 2$. Directed MPFR evaluation encloses its right-hand side in the interval recorded in the replication; its lower endpoint exceeds 6.37.

Finally, a smooth witness equal to g on all stopping faces and with $\sup_a \mathcal{R}_a v \leq e_+(s)$ gives $V(0, z) \leq v(0, z) + \int_0^1 e^{-.04s} e_+(s) ds$ by the stopped Itô identity. Take $z = (2, x)$, let $x \uparrow 2$, and use continuity of v and the established strict gap. The integrated uniform positive residual must exceed 6.37. A sequence with vanishing such residuals is excluded, without making any assertion that the economic control problem lacks a solution. *Q.E.D.*

PROOF OF THEOREM 6: Discounted Itô's formula, localization of the stochastic integral, and the bounded stopped-domain hypotheses give, for every admissible control a ,

$$J(t, z; a) - v(t, z) = \mathbb{E}_{t,z}^a \left[\int_t^\tau e^{-\rho(s-t)} \mathcal{R}_a v(s, Z_s) ds + e^{-\rho(\tau-t)} (g - v)(\tau, Z_\tau) \right].$$

The positive residual bound and the facewise upper discrepancy bounds imply

$$V - v \leq \int_t^T e^{-\rho(s-t)} e_+(s) ds + \sum_i \beta_i^+ \omega_i^*.$$

Apply the same identity to π with the opposite sign to obtain

$$v - J^\pi \leq \int_t^T e^{-\rho(s-t)} e_-(s) ds + \sum_i \beta_i^- \omega_i^\pi.$$

Adding proves the assertion. No common stopping time across two distinct controlled paths is assumed. For the proposed smooth hitting-probability bound w_i , optional stopping yields $\mathbb{E}[e^{-\rho(\tau-t)} w_i(\tau, Z_\tau)] \leq w_i(t, z)$; domination of the indicator on the stopping boundary then gives the stated probability bound. *Q.E.D.*

PROOF OF PROPOSITION 7: Write $Y = 2 - x > 0$ and $p = Yq$, where $q = (.15 + .65 \tanh N_p)/1.5$ is bounded. Until another exit,

$$dY = (c - .04)ds + Y\{.02 - .06xq\}ds - .2xqY dZ_s^x.$$

The bounded coefficients define a strictly positive stochastic exponential Φ . Variation of constants gives

$$Y_s = \Phi_s \left(Y_t + \int_t^s \Phi_r^{-1} (c_r - .04) dr \right) > 0,$$

since $c \geq .05$. The upper wealth face is therefore inaccessible from the interior. On the remaining state faces $B = 0$; at terminal time $1 - t = 0$. On the upper wealth face $B \geq 0$ and softplus is positive, so $v - g \geq 0$. Thus every upper-comparison trace discrepancy is zero, and the only potentially positive policy discrepancy has zero hitting probability. The same calculation applies to admissible bounded clipped head perturbations before the distance multiplication. *Q.E.D.*

PROOF OF THEOREM 7: Let $d_n = \epsilon_n/(1 - q)$. The resolvent of $I - qP_{\pi_n}$ is nonnegative and has row sums at most $(1 - q)^{-1}$, so $\|v_n - V^{\pi_n}\|_\infty \leq d_n$. Put $a_n = 2qd_n + \eta_n$. The contraction inequalities and approximate greedy inequality give

$$TV^{\pi_n} - T_{\pi_{n+1}} V^{\pi_n} \leq a_n \mathbf{1}.$$

Since $TV^{\pi_n} \geq V^{\pi_n}$, the same nonnegative resolvent gives $V^{\pi_{n+1}} \geq V^{\pi_n} - a_n \mathbf{1}/(1 - q)$. Therefore

$$\begin{aligned} V^* - V^{\pi_{n+1}} &= TV^* - T_{\pi_{n+1}} V^{\pi_{n+1}} \\ &\leq qE_n \mathbf{1} + a_n \mathbf{1} + qP_{\pi_{n+1}} (V^{\pi_n} - V^{\pi_{n+1}}) \\ &\leq \{qE_n + a_n/(1 - q)\} \mathbf{1}. \end{aligned}$$

Taking the supremum norm proves (35). The rate follows by summing the resulting geometric convolution. The comparison with a continuous-time problem is the triangle inequality after separately bounding the two transfer terms; it does not follow from the finite Markov contraction alone. *Q.E.D.*

APPENDIX F: PROOF OF THE STATE-GLOBAL INVENTORY CERTIFICATE

PROOF OF THEOREM 11: The scalar backward Riccati solutions exist on the finite horizon, remain at least .5, and are bounded above. Indeed, in remaining time their drift is $2a_i p - p^2 + q_i$, positive at .5 and negative above its positive equilibrium. The neural gains also remain at least .5 by construction. Let $d_i = \hat{p}_i - p_i^*$ and let R_i be the neural Riccati residual. Then

$$d'_i = (\hat{p}_i + p_i^* - 2a_i)d_i + R_i, \quad d_i(1) = 0.$$

The coefficient is at least $\kappa_i = 1 - 2a_i$. Integrating backward therefore yields $|d_i(t)| \leq \varepsilon_i(1 - e^{-\kappa_i(1-t)})/\kappa_i \leq D$.

The quadratic optimal value has matrix $P^* = p_\perp^*(I - M) + p_m^*M$ and a scalar noise term. Completing the square in its generator and applying Itô's formula gives the exact identity

$$\mathcal{J}(0, x; \hat{a}) - \mathcal{V}(0, x) = \mathbb{E} \int_0^1 |\{\hat{P}(s) - P^*(s)\}X_s|^2 ds.$$

The two projectors are orthogonal, so the integrand is bounded by $D^2|X_s|^2$. Both eigenvalues of $A - \hat{P}$ are at most $-.4$. Itô's formula for $|X_s|^2$ then gives

$$\frac{d}{ds} \mathbb{E}|X_s|^2 \leq -.8\mathbb{E}|X_s|^2 + .01d.$$

Gronwall's inequality and $|x|^2/d \leq 1$ imply $d^{-1}\mathbb{E}|X_s|^2 \leq e^{-.8s} + (.01/.8)(1 - e^{-.8s})$. Integration proves (61). The bounded deterministic gains give finite second moments and justify localization and removal of the stopping sequence in both Itô identities. No bounded-state truncation or simulated tail probability is used. *Q.E.D.*

APPENDIX G: PROOF OF THE COMPENSATION BOUND

PROOF OF PROPOSITION 13: Holding the specified paths fixed, the derivative of consumption utility with respect to the top-up is

$$\frac{\partial}{\partial \lambda} \frac{\{(1 + \lambda)c\}^{1-u}}{1-u} = c^{1-u}(1 + \lambda)^{-u} \geq (1 + \lambda)^{-2.8}.$$

The inequality uses $c \leq 1$, $u > 1$, $u \leq 2.8$, and $\lambda \geq 0$. Integrating from zero to λ , then over the discounted alive-time measure, bounds the welfare gain below by $A\{1 - (1 + \lambda)^{-1.8}\}/1.8$. Solving for a gain of ϵ proves (63). For the time-control library, wealth feasibility excludes a wealth exit, and the retained drift bounds and Brownian maximal inequality give the displayed preference-exit probability. Surviving to terminal time alone yields the stated lower bound on discounted alive time. This argument requires no extrapolation of the certified optimal value to an altered wealth state. *Q.E.D.*

APPENDIX H: PROOFS FOR THE CERTIFICATE-ALIGNED ALGORITHM

PROOF OF THEOREM 8: For a fixed slab and sign, write $A = \max\{0, \max_i a_i\}$ and $B = \max\{0, \max_i b_i\}$. The definition of d gives $b_i \leq \max\{a_i, 0\} + d \leq A + d$, hence $B \leq A + d$. For $K = M + 1$ logits including zero, factor $e^{A/\tau}$ out of their exponential sum. The remaining sum lies in $[1, K]$. It follows that

$$A \leq \tau \log(1 + \sum_i e^{a_i/\tau}) \leq A + \tau \log K.$$

Multiply by the positive discounted slab mass, sum over the two signs and all slabs, and apply Theorem 6. This proves (38); taking $a = b$ and $d = 0$ proves (39). A nonzero independently certified trace budget can be added to both policy-loss bounds without changing the argument.

For the implementation, every stored logit is interpreted as its exact dyadic value. The maximum used to stabilize the logarithm is also an exact stored value. Directed subtraction, exponentiation, pairwise summation, logarithm, multiplication, and discounted-mass evaluation enclose the smooth objective. Directed subtraction of each nonnegative proposal logit from its retained certified endpoint encloses the correction. Its construction is valid regardless of the source of the difference between the two computational graphs. An incomplete cover would invalidate the initial use of Theorem 6; the implementation rejects that case rather than increasing a numerical tolerance. *Q.E.D.*

PROOF OF PROPOSITION 8: The initial accepted certificate is valid by independent verification. At each subsequent checkpoint, rejection leaves the incumbent unchanged, and acceptance replaces it by a valid strictly smaller bound and its associated policy. Induction proves the first assertion.

For the second assertion, consider a deterministic unit-horizon problem with discount $\rho \geq 0$, zero terminal reward, and two actions with constant rewards 0 and -1 . Put $C_\rho(h) = \int_0^h e^{-\rho s} ds$ and $v_a(t) = aC_\rho(1-t)$. Then $v'_a - \rho v_a = -a$, so the optimal residual is $-a$ and the residual of an action with reward r is $r - a$. The signed certificate is

$$C_\rho(1)\{(-a)_+ + (a-r)_+\}.$$

The optimal action $r = 0$ paired with $a = -10$ has certificate $10C_\rho(1)$ and true regret zero. The worse action $r = -1$ paired with $a = -1$ has certificate $C_\rho(1)$ and true regret $C_\rho(1) > 0$. Both witnesses match terminal reward and both certificates are valid. Strict certificate acceptance therefore does not imply increasing policy values. This is an analytical counterexample to the inference, not a simulated observation about the original economy. *Q.E.D.*

PROOF OF PROPOSITION 9: Let $A_n = \|V_n^* - v_n\|_\infty$ and $P_n = \|V_n^\pi - v_n\|_\infty$. The evaluation and improvement inequalities imply

$$\|v_n - T_n v_{n+1}\|_\infty \leq \epsilon_n + \eta_n.$$

The Lipschitz inequalities for T_n and $T_n^{\pi_n}$ give

$$A_n \leq qA_{n+1} + \epsilon_n + \eta_n, \quad P_n \leq qP_{n+1} + \epsilon_n,$$

with $A_N, P_N \leq b$. Backward iteration and the triangle inequality give (40). Finally, $C_\rho(h) \sum_{n=0}^{N-1} e^{-\rho nh} = C_\rho(Nh)$, including its continuous extension at $\rho = 0$. The bound concerns this finite-horizon Bellman problem. Any use for a different continuous-time problem still requires a transfer theorem or a direct continuous-time certificate. *Q.E.D.*

PROOF OF PROPOSITION 12: The upper half of the verification argument gives

$$V_2(0, 2, x) \leq v(0, 2, x) + \int_0^1 e^{-.04s} e_+(s) ds$$

for every interior x . The upper trace inequalities remove any adverse settlement discrepancy. Let $x \uparrow 2$, use continuity of the fixed witness and Proposition 6, and subtract $g(0, 2, 2)$. The result is $\Delta \leq h_v + \int e^{-.04s} e_+(s) ds$. Since the integral is nonnegative, (51) follows. No policy payoff enters this lower bound; it is a statement about the residual needed to support that particular witness. Q.E.D.

APPENDIX I: PROOF FOR NONLINEAR VECTOR-STATE CONTROL

PROOF OF THEOREM 9: Let D be the cyclic first-difference matrix, so $(Dz)_i = z_i - z_{i+1}$ and $\|D\|_2 \leq 2$. Differentiating (54) gives

$$\nabla^2 \Phi(z) = \text{diag}(q_i + \frac{1}{2} \text{sech}^2 z_i) + D^\top \text{diag}(\frac{3}{10} \text{sech}^2 (Dz)_i) D.$$

All terms are positive semidefinite. Since $1 \leq q_i \leq 6/5$ and $\text{sech}^2 \leq 1$, $\nabla^2 \Phi(z) \preceq (6/5 + 1/2 + (3/10)4)I = (29/10)I$ globally.

Stack the H propagated states. Their derivative with respect to U is a block lower-triangular convolution operator B with coefficients $1, 3/5, (3/5)^2, \dots$. If S is the finite shift operator, then $B = \sum_{j=0}^{H-1} (3/5)^j S^j \otimes I_d$, so $\|B\|_2 \leq \sum_j (3/5)^j \leq 5/2$. Therefore

$$\nabla_U^2 F_x(U) = 4I + B^\top \text{diag}(\nabla^2 \Phi(z_{h+1})) B,$$

and its spectrum lies in $[4, 4 + (25/4)(29/10)] = [4, 177/8]$. This also establishes coercivity, existence, and uniqueness of the minimizer for every x .

Let $m = 4$, $L = 177/8$, and $\alpha = 2/(m + L) = 16/209$. With $g_k = \nabla F_x(U_k)$, the fundamental theorem of calculus yields

$$g_{k+1} = \left(I - \alpha \int_0^1 \nabla^2 F_x(U_k - t\alpha g_k) dt \right) g_k.$$

The integral is symmetric with spectrum in $[m, L]$. Consequently $\|g_{k+1}\|_2 \leq (L - m)/(L + m) \|g_k\|_2 = (145/209) \|g_k\|_2$. Strong convexity, followed by minimization of its quadratic lower tangent, gives $F_x(U) - \inf F_x \leq \|\nabla F_x(U)\|_2^2 / (2m) = \|\nabla F_x(U)\|_2^2 / 8$.

It remains to bound the initial gradient uniformly in the state cube and in the neural weights. The output constraint gives $|u_{0,hi}| \leq 1/4$, and $|b_{hi}| \leq .03$. State propagation therefore implies

$$|z_{hi}| \leq 1 + \frac{.25 + .03}{1 - .6} = 1.7.$$

Each coordinate of $\nabla \Phi$ has absolute value at most $(6/5)(1.7) + 1/2 + 2(3/10) = 3.14$. The backward adjoint is a geometrically weighted sum of such gradients. Hence every coordinate of $\nabla_U F_x(U_0)$ has absolute value at most $4(1/4) + 3.14/(1 - .6) = 8.85$. There are Hd coordinates, so $\|g_0\|_2^2 \leq Hd(8.85)^2$. Combining the inequalities proves (57) and solving it for k gives the work allocation.

Forward propagation, cyclic neighbor differences, and backward adjoints each require $O(Hd)$ operations; no dense Hessian need be formed. Directed verification of a stored plan replaces its

unknown exact gradient norm by an upper enclosure. It certifies that stored plan at that stored initial vector even when its floating-point optimization trajectory differs from the exact-real iteration. It does not, by itself, certify the floating-point neural/correction map simultaneously at every state. Q.E.D.

APPENDIX J: PROOF OF BUDGET-FEASIBLE COMPENSATION

PROOF OF PROPOSITION 14: Couple the original and compensated policies with the same preference Brownian motion. Their preference processes and adjustment costs coincide. Their portfolios are both zero. The wealth difference before stopping is

$$z(s) = \Delta x e^{.02s} - \frac{d_c}{.02} (e^{.02s} - 1) = \Delta x \frac{e^{.02} - e^{.02s}}{e^{.02} - 1} \geq 0,$$

with $z(1) = 0$. The verified wealth and consumption conditions make both policies admissible and exclude wealth stopping. Thus they have the same preference or terminal stopping time. At preference stopping, the wealth-dependent settlement increment is $.1 \log((X_s + z(s))/X_s) \geq 0$; at time one it is zero.

For any preference $u \in [1.2, 2.8]$, marginal utility is c^{-u} . Along the compensated consumption segment, $c \leq .8 < 1$, so $c^{-u} \geq .8^{-1.2}$. The mean value theorem therefore bounds the instantaneous utility increment below by $d_c \cdot 8^{-1.2}$. The survival probability at every time is at least $1 - p_e$. Integrating with discount and adding the nonnegative settlement change proves (64). The library's uniform policy-loss inequality then gives the concluding comparison with the original optimal value. The proof neither redefines that value nor asserts that the sufficient compensation is the minimum possible one. Q.E.D.

APPENDIX K: PROOFS FOR POLICY-SENSITIVE CONTINUATION AND LEARNED WORK

K.1. Separated acceptance and regret contraction

PROOF OF PROPOSITION 3: The enclosure inequalities give $J_{\pi'} - J_{\pi} \geq L_{\pi'} - U_{\pi}$. Moreover, $\bar{V} - L_{\pi} \geq V - J_{\pi}$. Under (25), subtracting the resulting gain inequality from $V - J_{\pi}$ proves the contraction. Under the last hypotheses, write $g = V - J_{\pi'} \geq 0$ and $a = J_{\pi'} - J_{\pi} \geq 0$. Then $g/(g+a) \leq g/(g+d) \leq B/(B+d)$. Since $d > 0$, the denominator is positive even when the new policy is optimal. These arguments apply separately at each state and therefore uniformly wherever the hypotheses are uniform. Q.E.D.

K.2. A difference-sensitive state modulus

PROOF OF THEOREM 5: First consider the unstopped reference functional used by the independent evaluator. At time s , couple the two preference inputs by the same standard normal variable Z :

$$U_s^i = u + m_i(s) + .05\sqrt{s}Z, \quad m_i(s) = \int_0^s \theta_v^i dv, \quad i \in \{a, b\}.$$

For the reference functional, clip the preference input to $[1.2, 2.8]$ in the running utility. Retain the analytically integrated terminal quadratic. This clipping does not change the original running payoff before a preference exit. The stopping allowance stated below controls the difference from the original stopped functional.

Write $f(u, c) = c^{1-u}/(1-u)$ and $r = u - 1$. Direct differentiation gives

$$|f_{uu}| = \frac{e^{-r \log c}}{r^3} \{(r \log c + 1)^2 + 1\},$$

$$|f_{uc}| = (-\log c)e^{-u \log c}, \quad |f_{cc}| = ue^{-(u+1) \log c}.$$

On the larger clipped rectangle, $|f_u| = e^{-r \log c}(1 + r \log c)/r^2$ and $|f_c| = e^{-u \log c}; 1 + r \log c$ is positive there. The code evaluates enclosing intervals for these formulas, not derivatives estimated from network outputs.

On $|Z| \leq 4$, all four coupled inputs, including those at z_* , are in $[1.78, 2.42] \times [.7, .8]$. This follows from $u \in [1.98, 2.02]$, $0 \leq m_i(s) \leq .2$, and $.05\sqrt{s}|Z| \leq .2$. No clipping occurs on this event. The difference between the two policies' preference derivatives is bounded by $H_{uu}d_m + H_{uc}d_c$; the corresponding consumption derivative difference is bounded by $H_{uc}d_m + H_{cc}d_c$. Integrate along the straight segment from z_* to z . Its preference displacement is at most .02, and its common consumption displacement is at most $.01/Q_r$. The resulting bound on the change of the payoff difference is the central term inside the brackets of (28).

On the complementary event, clipping is 1-Lipschitz. For each of the two initial states the difference of the two clipped running utilities has absolute value at most $A_u d_m + A_c d_c$. Each clipped running utility is negative and has absolute value less than 6: on this rectangle, $e^{-r \log c}/r$ decreases in r , and its largest value is at $r = .2, c = .7$, where it is less than 6. The oscillation of the difference between two initial states is consequently at most D_4 . Mills' inequality gives

$$\Pr(|Z| > 4) \leq 2 \frac{e^{-8}}{4\sqrt{2\pi}} = P_4.$$

This is a one-time normal-tail bound inside the time integral, not a bound on the maximum of a Brownian path. Integrating the two running-payoff bounds with the discount gives the first term in (28). Preference-adjustment expenditure is independent of the initial state and cancels from the state difference.

The common wealth shift implies

$$X_1^i = e^{.02} \left\{ x - \int_0^1 e^{-.02s} \left(c_s^i + \frac{x - 1.25}{Q_r} \right) ds \right\} = e^{.02} \left\{ 1.25 - \int_0^1 e^{-.02s} c_s^i ds \right\}.$$

Thus each policy's terminal wealth is independent of x , even when the stored dyadic controls do not satisfy the ideal budget identity exactly. Their log-settlement difference has zero initial-state variation. The difference of their expected terminal preference settlements has initial-preference derivative $-.04(m_b(1) - m_a(1))$. Its discounted variation is at most $.04e^{-\rho}(.02)d_T$.

Finally, both policies have strictly inward wealth drift and terminal wealth above .5, verified over K . Their only possible premature exit is in preference. For $0 \leq \theta \leq .2$, the distance of the deterministic preference path from either boundary is at least .58 on K . The exponential Brownian maximal bound therefore gives

$$p_e = 2e^{-.58^2/(2(.05)^2)}.$$

The independent original-payoff argument bounds the discrepancy between the clipped reference functional and the stopped value by

$$\epsilon_{\text{stop}} = 15p_e + .02\sqrt{\{.22^4 + 6(.22)^2(.0025) + 3(.0025)^2\}p_e}.$$

The second term uses the fourth moment and Cauchy–Schwarz for the terminal quadratic, rather than treating that unbounded random variable as uniformly bounded. There are two policies at two initial states, so the transfer calculation pays four such allowances. Adding these terms proves (29). The central quadrature enclosures are then combined with the directed modulus by interval subtraction and addition. No simulation approximation to an exit time enters this argument. *Q.E.D.*

K.3. Four-corner optimality for the new actors

The exact lower functional in Lemma 1 depends on feasibility, the budget-preserving shift, and the original utility, not on the algorithm that generated the controls. Each compiled neural policy passes those same conditions. Its four independently evaluated lower endpoints are bounds for the same concave covered functional after uniform quadrature, tail, and stopping allowances. The two inherited dual upper functions are convex in initial state and are valid at their respective prices. Convexity of V in k makes their chord a valid optimal upper comparison at $k = 2$. The argument of Proposition 10 therefore bounds regret over all of K by the largest of the four new corner gaps. This application does not interpolate raw Monte Carlo values and does not infer concavity of an arbitrarily fitted neural critic.

K.4. A constructive continuation trace

PROOF OF PROPOSITION 5: Fix the stored no-transfer time controls. The wealth solution is deterministic, affine in initial wealth, and continuous in that initial condition. For every $x \in [1.25, 2)$, its drift satisfies $.02X_s - c_s < 0$ and its terminal wealth exceeds .5 by the directed budget check. Thus it does not hit a wealth boundary. Preference exit depends on u , the Brownian motion, and the same fixed θ , but not on initial wealth. The stopping time is therefore common as $x \uparrow 2$. On this range the running payoff is bounded before stopping, and the wealth component of settlement is bounded and continuous. Dominated convergence proves existence of the interior limit and identifies its payoff by inserting the limiting wealth path into the same preference-stopped expectation.

The independent quadrature evaluates the unclipped Gaussian representation with its normal-tail and stopping allowances. Those allowances continue to hold for this fixed control context and the limiting wealth path: consumption and preference ranges are unchanged, wealth remains in $[\cdot 5, 2]$, and the sole premature exit is in preference. The evaluator’s no-transfer input adapter checks these hypotheses; it does not invoke the usual K -dependent wealth shift at $x = 2$.

For uniformity in u , consider the covered reference payoff minus $g(0, u, 2)$. Discounting leaves a positive terminal-minus-initial quadratic curvature of $.04(1 - e^{-.04})$; those quadratic terms do not cancel. The running utility supplies a larger negative curvature. Indeed, on the covered region $r = u - 1 \leq 1.8$, $c \leq .8$, so

$$-f_{uu} = \frac{e^{-r \log c}}{r^3} \{(r \log c + 1)^2 + 1\} \geq \frac{125}{729}.$$

The covered normal mass is at least $1 - 2e^{-32}$, and $e^{-.04s} \geq e^{-.04}$. Thus the second derivative of the covered payoff minus initial settlement is at most

$$-\frac{125}{729}e^{-.04}(1 - 2e^{-32}) + .04(1 - e^{-.04}) < 0.$$

Expenditure and terminal log wealth are independent of u . Subtracting the common tail, quadrature, and stopping allowances preserves this strict concavity. Its minimum on $[1.98, 2.02]$ is

bounded below by the smaller endpoint enclosure. Directed evaluation of both endpoints for all five fixed policies gives the stated minimum; the additional center calculation is a diagnostic rather than the basis of uniformity.

The process defined by immediate settlement when initialized exactly at $x = 2$ is a different initial-value convention. The proposition concerns only the interior limit and makes no change to that boundary convention. *Q.E.D.*

K.5. Budget-feasible neural compensation

PROOF OF PROPOSITION 4: Let $Q_r(s) = \int_0^s e^{-.02v} dv$. For the paired original and compensated plans the difference in wealth is

$$\tilde{X}_s - X_s = e^{.02s} \delta \left(1 - \frac{Q_r(s)}{Q_r(1)} \right) \geq 0, \quad \tilde{X}_1 - X_1 = 0.$$

The independently checked consumption ranges lie inside $[.7, .8]$ even after both the K shift and the compensating increment. Both wealth paths are strictly decreasing and remain above .5; the compensated initial wealth is below 2. Neither can exit in wealth. Their preference paths, hence preference exit times, coincide. Logarithmic wealth settlement cannot fall at that time. Preference expenditure and preference settlement are unchanged.

Before stopping, $f_c = c^{-u} \geq (.8)^{-1.2}$. Hence the running utility gain is at least $(\delta/Q_r)(.8)^{-1.2}$ per unit of discounted time. Since the probability of a premature preference exit is at most p_e , the expected discounted time before stopping is at least $C_\rho(1)(1 - p_e)$. The directed lower bound on the gain is larger than .0119058, whereas every new rectangular regret upper bound is at most 0.011816625. Subtracting regret proves the last inequality in the proposition. This construction changes the feasible consumption policy with wealth; it is not a claim about the minimum wealth increment for a fixed policy. *Q.E.D.*

K.6. Learned-gradient work

PROOF OF THEOREM 10: For fixed x , let $g = \nabla F_x(U)$ and $\eta = 2/(m + L)$. The integral Hessian identity gives

$$\nabla F_x(U - \eta g) = \left[I - \eta \int_0^1 \nabla^2 F_x(U - s\eta g) ds \right] g.$$

The integral matrix is symmetric with spectrum in $[m, L]$. The operator norm of the bracket is at most $(L - m)/(L + m) = 145/209$. Iteration and the strong-convexity bound $F_x(U) - F_x^* \leq \|\nabla F_x(U)\|^2/(2m)$ prove (58). In particular the initial gradient, not merely bounded network outputs, determines this guarantee.

For the state extension, write the stacked inventory trajectory as $AU + Sx$ plus its fixed offset. The model's geometric propagation gives $\|A\|_2 \leq 5/2$ and $\|S\|_2 \leq 3/4$. The nonlinear inventory potential has Hessian norm at most $29/10$; the quadratic control cost has no state cross derivative. Consequently

$$\|D_x \nabla_U F_x(U)\|_2 \leq (5/2)(29/10)(3/4) = 87/16.$$

For any covering center x_i ,

$$\begin{aligned} \|\nabla F_x(f_\theta(x))\|_2 &\leq \|\nabla F_{x_i}(f_\theta(x_i))\|_2 + L\|f_\theta(x) - f_\theta(x_i)\|_2 + (87/16)\|x - x_i\|_2 \\ &\leq G_\theta(x_i) + (LK_\theta + 87/16)h. \end{aligned}$$

A tanh activation is 1-Lipschitz. Multiplying enclosing matrix-norm bounds, including the output scale, gives a valid K_θ ; Frobenius norms can replace spectral norms conservatively. This statement requires a genuine cover of the asserted set. A finite query maximum instead corresponds to $h = 0$ on that finite set alone. *Q.E.D.*

REFERENCES

- Barles, G., and P. E. Souganidis (1991): “Convergence of Approximation Schemes for Fully Nonlinear Second Order Equations,” *Asymptotic Analysis*, 4, 271–283. []
- Brown, D. B., J. E. Smith, and P. Sun (2010): “Information Relaxations and Duality in Stochastic Dynamic Programs,” *Operations Research*, 58, 785–801. []
- Brown, D. B., and J. E. Smith (2011): “Dynamic Portfolio Optimization With Transaction Costs: Heuristics and Dual Bounds,” *Management Science*, 57, 1752–1770. []
- Brumm, J., and S. Scheidegger (2017): “Using Adaptive Sparse Grids to Solve High-Dimensional Dynamic Models,” *Econometrica*, 85, 1575–1612. []
- Cai, W., S. Fang, and T. Zhou (2025): “SOC-MartNet: A Martingale Neural Network for the Hamilton–Jacobi–Bellman Equation without Explicit $\inf_{u \in U} H$ in Stochastic Optimal Controls,” *SIAM Journal on Scientific Computing*, 47, C795–C819. []
- Han, J., and W. E (2016): “Deep Learning Approximation for Stochastic Control Problems,” arXiv:1611.07422. []
- Han, J., A. Jentzen, and W. E (2018): “Solving High-Dimensional Partial Differential Equations Using Deep Learning,” *Proceedings of the National Academy of Sciences*, 115, 8505–8510. []
- Judd, K. L. (1998): *Numerical Methods in Economics*. Cambridge, MA: MIT Press. []
- Kushner, H. J., and P. Dupuis (2001): *Numerical Methods for Stochastic Control Problems in Continuous Time*, second edition. New York: Springer. []
- National Institute of Standards and Technology (2026): *Digital Library of Mathematical Functions*, Section 3.5, “Quadrature,” online reference consulted September 22, 2026. []
- Fousse, L., G. Hanrot, V. Lefèvre, P. Pélissier, and P. Zimmermann (2007): “MPFR: A Multiple-Precision Binary Floating-Point Library With Correct Rounding,” *ACM Transactions on Mathematical Software*, 33(2), Article 13. []